

Interbank Markets, Foreign Banks, and Monetary Policy Under a Floor System*

Duhyeong Kim[†]
Kent State University

December 16, 2025

Abstract

This paper investigates the role of foreign banks in interbank markets and its implications for U.S. monetary policy under a floor system. We develop a two-country, two-sector banking model to illustrate the behavior of domestic banks, U.S. branches of foreign banks, and other financial institutions in interbank markets. We find that a central bank's balance sheet expansion can increase foreign banks' reserve holdings and their activity in interbank markets, while reducing welfare for both domestic and foreign consumers. In contrast, introducing an overnight reverse repo facility reduces foreign banks' interbank market participation and improves welfare globally.

Keywords: Interbank markets; Floor system; Interest on reserves; Foreign banks; Overnight reverse repurchase agreement facility

JEL Codes: E4; E5

*I would like to thank Qingqing Cao, Jonathan Chiu, Sergio Ocampo Diaz, Heon Lee, Steve Williamson, Fengfan Xiang, and participants at Yonsei University, the 2024 Canadian Economic Association conference, the 2024 Summer Workshop on Money, Banking, Payments, and Finance at the Federal Reserve Board, the 2024 Northeast Ohio Economics Workshop at the Federal Reserve Bank of Cleveland, and the 2025 Southern Economics Association conference for helpful discussions, comments, and suggestions. I also thank the editor, Tommaso Monacelli, and anonymous referees for their valuable comments and suggestions. All remaining errors are mine.

[†]Email address: dkim33@kent.edu

1 Introduction

In the aftermath of the 2007-2008 global financial crisis, the Federal Reserve adopted a new operating framework for monetary policy, known as the *floor system*. Under this system, the Federal Reserve administratively sets the interest on reserve balances (IORB) rate and adjusts the size of its balance sheet through large-scale asset purchases and sales. This new environment has enabled U.S. branches and agencies of foreign banks (“foreign banks”, hereafter) to significantly expand their holdings of reserve balances and has made them key players in the transmission of U.S. monetary policy.

Foreign banks became central to monetary policy transmission for two main reasons. First, under the floor system, foreign banks emerged as major borrowers in the federal funds (fed funds) market, while participation by domestic banks declined substantially. Because the IORB rate did not turn out to serve as a floor for the fed funds rate, foreign banks were able to earn essentially risk-free arbitrage profits by borrowing in the fed funds market and holding the funds as reserves. As a result, movements in the fed funds rate became increasingly dependent on the activities of foreign banks. Second, during periods of the Federal Reserve’s balance sheet expansion through large-scale asset purchases, foreign banks absorbed the majority of the increase in reserves. Between Q1 2008 and Q3 2014, foreign banks’ share of total reserves rose from 17.8 percent to 50.2 percent, and their reserves-to-assets ratio increased sharply from 4.1 percent to 56.2 percent.¹

How do changes in monetary policy—specifically, changes in the IORB rate and the size of the Federal Reserve’s balance sheet—affect other interest rates and economic welfare in this new environment? Given the role of foreign banks in allocating resources across countries, what are the domestic and global implications of U.S. monetary policy? To address these questions, we develop a two-country, two-banking-sector general equilibrium model in which *retail banks* and *unconventional* (non-retail) *banks* interact through an interbank market.

To replicate the landscape of the fed funds market, we introduce two institutional differences among banks. First, a legal restriction permits only retail banks to earn interest on reserve balances. Despite this restriction, arbitrage would still act to close the spread between the IORB rate and the interbank lending rate. If the IORB rate exceeds the interbank rate, retail banks would competitively borrow funds to earn the risk-free interest margin, thereby eliminating the spread. However, this implication is inconsistent with the persistent spread observed in the left panel of [Figure 1](#), which shows that the IORB rate has generally exceeded the federal funds rate since 2008.

To endogenize the interest rate spread, we introduce capital requirements for retail banks. This assumption effectively generates “balance sheet costs” that retail banks incur when expanding their balance sheets, limiting the scope of their arbitrage activities.² To capture heterogeneity in balance sheet costs between retail banks based in the *home* country (*home banks*) and those based in the *foreign* country (*foreign banks*), we assume that home banks, representing U.S. banks, face tighter

¹During this period, U.S. banks’ reserves-to-assets ratio increased only modestly, from 2.7 percent to 11.9 percent.

²Notice that, in practice, balance sheet costs include not only binding capital requirements but also deposit insurance premiums ([Bech and Klee, 2011](#); [Afonso, Entz, and LeSueur, 2013](#); [Ennis, 2018](#); [Williamson, 2019](#)).

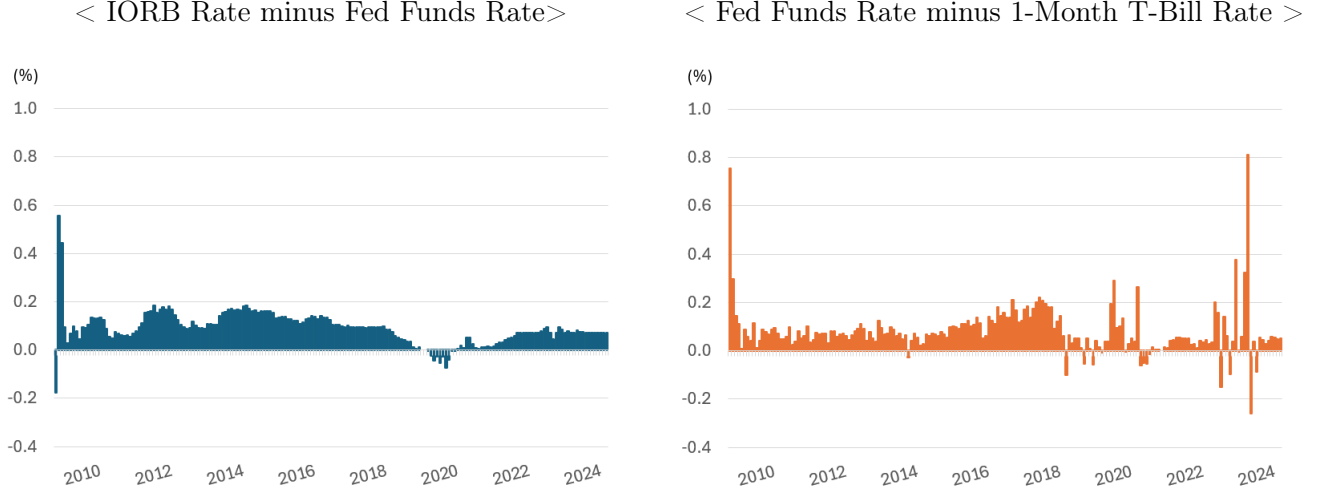


Figure 1: Interest rate spread among short-term interest rates (monthly average)

Source: Federal Reserve Economic Data

capital constraints. This difference in balance sheet costs helps explain why foreign banks become active borrowers in the interbank market, while home banks remain largely inactive.³

In the model developed here, all agents face a limited commitment problem, meaning they can default on their liabilities. Thus, when retail and unconventional banks create deposit liabilities, their assets must serve as collateral to prevent potential defaults. This introduces an additional type of balance sheet constraint—*collateral constraints*. We assume that capital constraints are tighter than collateral constraints for home banks, whereas collateral constraints are tighter for foreign banks. In equilibrium, a shortage of available collateral leads to binding balance sheet constraints: capital constraints for home banks and collateral constraints for foreign and unconventional banks.

Lastly, we assume that banks provide services to different sets of customers depending on their type. Retail banks offer traditional banking services to domestic consumers; that is, retail bank depositors can withdraw domestic currency from their deposit accounts or use deposit claims as a means of payment. Unconventional banks provide financial services to both domestic and foreign customers. In addition, some unconventional bank depositors prefer to use government bonds for financial transactions (e.g., repurchase agreements). This assumption allows us to generate a spread between the interest rate on government bonds and the interbank lending rate. As illustrated in the right panel of Figure 1, for most of the period since 2008, the one-month Treasury bill rate has been lower than the fed funds rate.

³Specifically, U.S. banks must adhere to leverage ratios based on daily or weekly averages, whereas some foreign banks are required to meet leverage ratios only at the end of each reporting period. The smaller burden of balance sheet costs allows foreign banks to exploit arbitrage opportunities in the fed funds market by increasing their reserve holdings. Another reason foreign banks face lower balance sheet costs is that many foreign banks operating in the United States are not insured by the Federal Deposit Insurance Corporation (FDIC); therefore, their insurance premiums do not increase with the size of their balance sheets (Afonso, Entz, and LeSueur, 2013; Kreicher, McCauley, and McGuire, 2014; Ennis and Wolman, 2015; Banegas and Tase, 2022).

Our model first demonstrates that, under a floor system, balance sheet constraints can generate an interest rate spread between reserves and government bonds. Specifically, when the supply of government bonds is sufficiently low relative to reserves, the collateral constraints faced by unconventional banks are tight, as these banks do not have access to reserves. This implies that, relative to reserves, the liquidity premium (or convenience yield) on government bonds is higher, leading to a higher bond price and a lower interest rate.⁴ In this scenario, unconventional banks competitively seek alternative interest-bearing assets to better serve their depositors, which drives the interbank lending rate below the IORB rate. Foreign banks are then incentivized to engage in arbitrage, while home banks remain inactive due to their high balance sheet costs. This generates an active interbank market in which funds flow from unconventional banks to foreign banks.

When both government bonds and interbank loans serve as collateral at the margin, the interest rates on these two assets are identical in equilibrium. However, when government bonds become extremely scarce, they are used exclusively as a means of payment rather than as collateral. In this case, the market for government bonds is effectively segmented from the interbank market. Reflecting a higher liquidity premium, the interest rate on government bonds then falls below the interbank lending rate.

In our model, the *home* central bank's balance sheet expansion effectively reduces the supply of *home* government bonds in the financial system, as it involves exchanging reserves for bonds. Depending on the size of the central bank's balance sheet, different types of equilibria arise. When the balance sheet is sufficiently small, a *type-1 equilibrium* emerges, in which there is no interest rate spread between reserves and government bonds and no interbank market activity. When the balance sheet is sufficiently large, but not extremely large, a *type-2 equilibrium* occurs, in which the interest rate on government bonds falls below the IORB rate. In this case, the interbank market becomes active, and the interbank lending rate equals the interest rate on government bonds. When the balance sheet is extremely large, a *type-3 equilibrium* arises, in which the interest rate on government bonds is lower than the interbank lending rate, which in turn is lower than the IORB rate.

Having identified the conditions that generate active interbank transactions between foreign and unconventional banks, we use the model to analyze the effects of the *home* central bank's monetary policy. A reduction in the IORB rate leads to a substitution of currency for reserves. Consequently, the real quantity of currency increases, while the quantity of bank deposits declines. A lower interest rate on reserves also reduces the interest rates on other collateralizable assets and the redemption value of collateral, thereby tightening collateral constraints. This adversely affects the foreign and unconventional banking sectors and ultimately reduces welfare for foreign consumers.

The effects of balance sheet expansion depend on the type of equilibrium. In a type-2 equilibrium, where the central bank's balance sheet is sufficiently large, expansion can decrease global

⁴The way the model generates an interest rate spread is consistent with [Arrata, Nguyen, Rahmouni-Rousseau, and Vari \(2020\)](#), who find that a decrease in the stock of collateral lowered the "repo" rate below the IORB rate. Specifically, they show that a 1% purchase of an outstanding bond by the European Central Bank led to a decline in its repo rate of 0.78 basis points.

welfare. The central bank’s exchange of reserves for government bonds initially tightens the collateral constraints of unconventional banks that cannot hold reserves. The resulting increase in the liquidity premium on collateralizable assets eventually tightens the balance sheet constraints faced by other banks, leading to contractions across all banking sectors and a decline in welfare for both home and foreign consumers. Conversely, in a type-3 equilibrium, where the balance sheet is extremely large, expansion can have redistributive effects within the banking system. Because the market for home government bonds is effectively segmented from the markets for other assets, this intervention expands the home and foreign banking sectors by increasing their stock of collateral, while causing a contraction in the unconventional banking sector by reducing transactions involving government bonds.

Interestingly, even when the central bank’s balance sheet expansion increases the interest rate spread between interbank loans and reserves in a type-2 equilibrium, foreign consumers experience a decrease in welfare. Although foreign banks earn higher arbitrage profits by increasing their share of reserves and enjoying a higher interest margin—an effect that would normally enhance foreign consumers’ welfare—this benefit is outweighed by the welfare losses arising from tighter collateral constraints and the associated increase in balance sheet costs. As a result, foreign consumers’ welfare declines in equilibrium.

We also use the model to examine the implications of an overnight reverse repurchase agreement (ON-RRP) facility. This facility is designed to give the Federal Reserve better control over the fed funds rate by allowing non-bank financial institutions to hold interest-bearing central bank liabilities. Because foreign banks rely on interbank borrowing to engage in arbitrage, the facility is expected to affect them more strongly. In the model, introducing the ON-RRP facility—which replaces reserves with ON-RRPs—indeed reduces foreign banks’ share of reserves as well as arbitrage profits. However, unconventional banks’ holdings of ON-RRPs eventually relax not only their own collateral constraints but also balance sheet constraints across the financial system, allowing all banks to operate more efficiently despite fewer reserves. Thus, welfare rises for both home and foreign consumers.

Literature Review—This paper is related to several strands of the literature in monetary economics and banking. First, it contributes to the literature on the behavior of foreign banks and its implications for U.S. monetary policy. [Afonso, Entz, and LeSueur \(2013\)](#) document changes in the fed funds market under the floor system and emphasize the role of foreign banks. [Du \(2023\)](#) shows that short-term lending rates, such as the fed funds rate and the repo rate, are largely driven by foreign banks’ activities and that fluctuations in these rates depend critically on the supply of reserves and banks’ balance sheet constraints (see also [Anderson, Du, and Schlusche, 2025](#)). [Banegas and Tase \(2022\)](#) document that foreign banks engage in window dressing on reporting days by temporarily reducing reserve holdings by 18.3 percentage points relative to domestic banks.⁵ These

⁵Balance sheet costs diverged further after the introduction of the revised FDIC assessment rule, which applied only to U.S. banks. [Kreicher, McCauley, and McGuire \(2014\)](#) and [Ennis and Wolman \(2015\)](#) study the implications of the expansion of the FDIC assessment base for bank asset portfolios and find that foreign banks played a crucial

empirical findings suggest relatively low balance sheet costs faced by foreign banks and highlight their importance in the transmission of monetary policy.

On the theoretical side, [Stern \(2022\)](#) extends the framework of [Bianchi and Bigio \(2022\)](#) to incorporate foreign banks with lower balance sheet costs. She shows that foreign banks are more responsive to changes in the interest rate on reserves, highlighting their importance for the transmission of monetary policy. [Oguri and Pizzimenti \(2025\)](#) estimate heterogeneous reserve demand elasticities across banks and develop a partial equilibrium model of reserves, based on [Poole \(1968\)](#), to study the optimal supply of reserves. They find that the high demand elasticity of reserve demand by foreign banks has important implications for monetary policy, particularly during balance sheet contraction. With a different focus, our two-country general equilibrium model allows us to study how central bank balance sheet policies affect interbank markets, the allocation of reserves across banks, and global welfare.

Second, this paper relates to the literature that studies the implications of interbank markets for monetary policy in a closed economy. Seminal and recent contributions include [Poole \(1968\)](#), [Ennis and Keister \(2008\)](#), [Afonso and Lagos \(2015\)](#), [Armenter and Lester \(2017\)](#), [Afonso, Armenter, and Lester \(2019\)](#), [Williamson \(2019, 2023\)](#), [Bianchi and Bigio \(2022\)](#), [Afonso, Giannone, La Spada, and Williams \(2023b\)](#), and [Afonso, La Spada, Mertens, and Williams \(2023c\)](#). Unlike these studies, which abstract from the role of foreign banks, our paper focuses on the behavior of foreign banks and examines both the domestic and global implications for monetary policy.

Lastly, this paper contributes to the money-search literature that analyzes the international effects of monetary policies. Examples include [Matsuyama, Kiyotaki, and Matsui \(1993\)](#), [Wright and Trejos \(2001\)](#), [Head and Shi \(2003\)](#), [Gomis-Porqueras, Kam, and Lee \(2013\)](#), [Geromichalos and Simonovska \(2014\)](#), [Zhang \(2014\)](#), [Gomis-Porqueras, Kam, and Waller \(2017\)](#), [Herrenbrueck \(2017\)](#), [Geromichalos and Jung \(2018\)](#), [Lee and Jung \(2020\)](#), and [Kim \(2023\)](#). Among these, [Kim \(2023\)](#) is most closely related to our work, as it also develops a two-country framework to study how a central bank’s balance sheet expansion generates global spillover effects. [Kim \(2023\)](#) shows that, because short-term assets are more pledgeable as collateral than long-term assets, a central bank’s swaps of short-term government liabilities for long-term ones can increase the effective stock of collateral in the financial system and thereby improve global welfare. In that model, reserves and short-term government bonds are essentially perfect substitutes.

This paper departs from [Kim \(2023\)](#) by focusing on replicating the behavior of foreign banks in the U.S. interbank market, while abstracting from differences in the maturities of government liabilities. To this end, we introduce imperfect substitutability between reserves and government bonds through the presence of two banking sectors—retail and unconventional—and a legal restriction that allows only retail banks to earn interest on reserves.⁶ Additionally, we incorporate asymmetric

role in absorbing the increase in reserves after 2011. [Banegas and Tase \(2022\)](#) show that the revision of the FDIC rule led to a 4.5 percentage point increase in the reserves-to-assets ratio of foreign banks relative to domestic banks.

⁶Introducing the assumption of asymmetric balance sheet costs alone cannot generate interbank fund flows, and the resulting welfare implications would remain the same as in [Kim \(2023\)](#).

balance sheet costs between home and foreign banks to generate interbank transactions that resemble those observed in the fed funds market. As a result, our model delivers welfare implications that differ sharply from those in [Kim \(2023\)](#): a central bank’s balance sheet expansion—implemented through swaps of reserves for government bonds—can reduce global welfare by increasing the balance sheet costs of both retail and unconventional banks.

2 Evolution of the Federal Funds Market

The federal funds (fed funds) market is an unsecured, mostly overnight, funding market in which depository institutions and government-sponsored enterprises (GSEs) borrow and lend central bank reserves. Prior to the global financial crisis of 2007-2008, this market played a crucial role in redistributing reserve balances throughout the banking system. Institutions that needed funds for clearing, settlement, or to meet reserve requirements would borrow from those holding excess reserves. In this environment, the Federal Reserve intervened daily in the market for repurchase agreements to maintain its target for the fed funds rate.

In response to the global financial crisis, the Federal Reserve launched a large-scale asset purchase program to support the U.S. economy as the fed funds rate reached its effective lower bound. This program expanded the Fed’s balance sheet, resulting in a large supply of reserves in the banking system. This substantially reduced the need for institutions to borrow in the fed funds market for regular purposes. The Fed also began setting a positive interest on reserve balances (IORB) rate to encourage depository institutions to hold reserves in large quantities. Consequently, the incentives for these institutions to lend in the fed funds market—particularly at rates below the IORB rate—have diminished.⁷ This new policy framework, characterized by a large quantity of reserves and the IORB rate, is known as the *floor system*, because the IORB rate was expected to set a floor for the fed funds rate.

While the landscape of the fed funds market evolved under the floor system, foreign banks—U.S. branches and agencies of foreign banks—became the most active borrowers in the market. Since 2016, foreign banks have borrowed approximately 65-95 percent of fed funds.⁸ This activity has been accompanied by a significant increase in their reserve holdings. As shown in [Figure 2](#), foreign banks’ share of total reserves rose from 17.8 percent in the first quarter of 2008 to 50.2 percent in the third quarter of 2014.⁹ Moreover, over the same period, the ratio of reserves to total assets for foreign banks surged from 4.1 percent to 56.2 percent, whereas the ratio for domestic banks increased only from 2.7 percent to 11.9 percent.

⁷In fact, trading volume in the fed funds market has declined substantially since the Federal Reserve implemented these policies. Prior to the crisis, daily trading volume ranged from \$150 billion to \$175 billion. This volume fell to approximately \$60 billion to \$80 billion in the 2010s ([Afonso, Cisternas, Gowen, Miu, and Younger, 2023a](#)).

⁸Between Q2 2020 and Q2 2024, foreign banks borrowed 80-95 percent of total fed funds ([Anderson and Na, 2024](#)). [Figure 10](#) in Appendix A.1 shows the shares of fed funds borrowing and lending by institution type from 2016 to 2024.

⁹A similar pattern is observed during the Federal Reserve’s large-scale asset purchases in response to the COVID-19 pandemic. The share of reserves held by foreign banks initially declined from 34.2 percent in Q4 2019 to 24.2

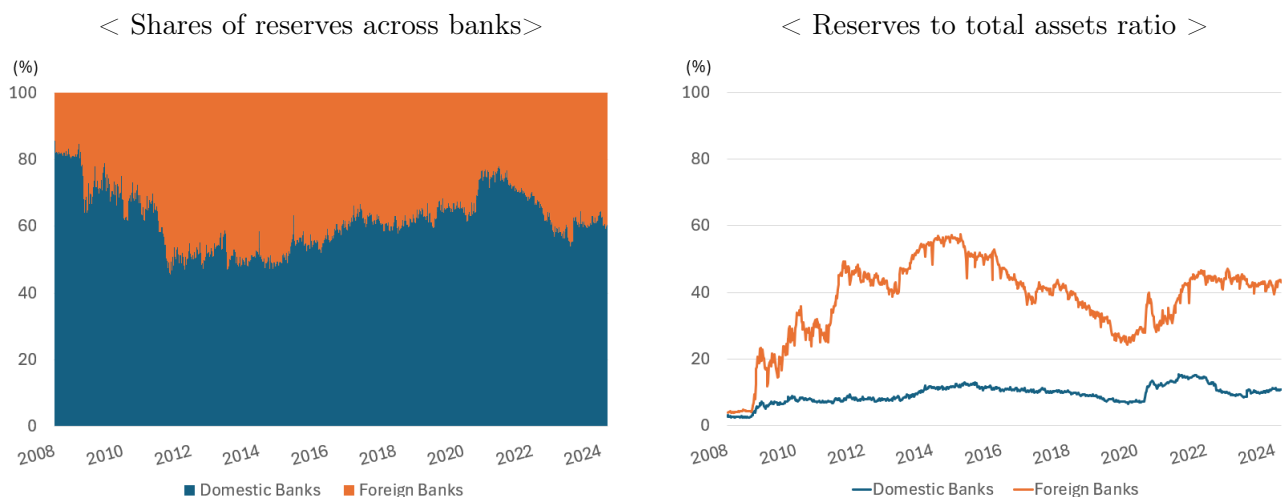


Figure 2: Shares of reserves across banks (left) and reserves to total assets ratio (right)

Source: Federal Reserve Board

The increase in the share of reserves held by foreign banks coincides with the Federal Reserve’s balance sheet expansion. [Figure 3](#) shows that foreign banks’ share of reserves was positively correlated with the size of the Fed’s balance sheet between January 2008 and December 2019. The figure also highlights a structural shift in this relationship that occurred in September 2013, when the Fed introduced another policy tool—the overnight reverse repurchase agreement (ON-RRP) facility. Specifically, the ON-RRP facility reduced the reserve holdings of foreign banks relative to those of domestic banks. This finding, together with differences in reserves-to-assets ratios between domestic and foreign banks, suggests that foreign banks are particularly responsive to the Fed’s new policies, such as the large-scale asset purchase program and the ON-RRP facility.¹⁰

The U.S. branches and agencies of foreign banks conduct a limited set of banking activities on behalf of their parent institutions. Branches are not permitted to accept retail deposits, while agencies can accept neither retail nor wholesale deposits from U.S. citizens.¹¹ This means that these institutions have increased their holdings of reserves for purposes other than clearing and settlement. Indeed, the fed funds rate has consistently remained below the IORB rate, creating arbitrage opportunities for foreign banks. The presence of this interest rate gap, along with the differing responses of domestic and foreign banks, raises the following questions. First, why do foreign banks not fully exploit these arbitrage opportunities, and why do domestic banks not actively borrow in the fed funds market? Second, which institutions lend in the fed funds market, and what motivates their lending behavior?

percent in Q3 2020, before rising to 42.6 percent by Q1 2023.

¹⁰[Oguri and Pizzimenti \(2025\)](#) also find that foreign banks absorb the majority of the increase in reserves when the Fed expands its balance sheet. Relatedly, [Stern \(2022\)](#) shows that foreign banks are more responsive to changes in the interest rate on reserves.

¹¹See [DiSalvo \(2019\)](#) for more information about how foreign banks operate in the United States.

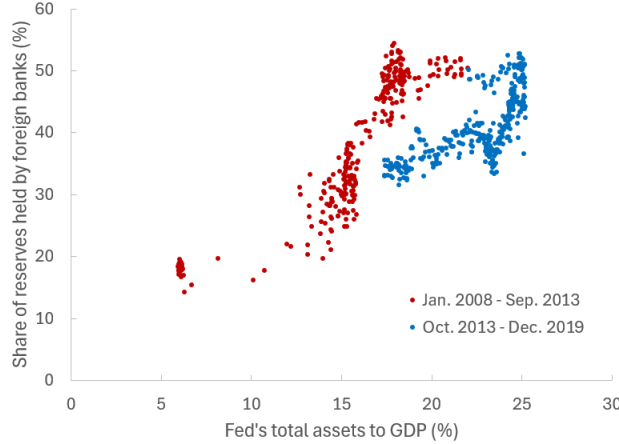


Figure 3: Size of the Fed’s balance sheet and the share of reserves held by foreign banks

Source: Federal Reserve Board

2.1 Domestic versus foreign banks

When the fed funds rate is below the IORB rate, banks can borrow in the fed funds market and deposit the funds at the Fed to earn an essentially risk-free interest margin. The persistent spread between the IORB rate and the fed funds rate therefore indicates the presence of substantial frictions that limit arbitrage. These frictions arise in part from the costs that banks incur when expanding their balance sheets. Such costs, commonly referred to as “balance sheet costs,” reflect deposit insurance premiums and binding capital requirements, such as leverage ratios (Bech and Klee, 2011; Afonso, Entz, and LeSueur, 2013; Ennis, 2018; Williamson, 2019). Because these costs render arbitrage activities unprofitable, the spread between the IORB rate and the fed funds rate persists.

However, balance sheet costs are heterogeneous across banks. For example, many foreign banks operating in the United States are not insured by the Federal Deposit Insurance Corporation (FDIC), and therefore their insurance premiums do not increase with the size of their balance sheets. In addition, cross-country differences in leverage ratio calculations make it less costly for foreign banks to expand their balance sheets. Specifically, U.S. banks must adhere to leverage ratios based on daily or weekly averages, whereas some foreign banks are required to meet leverage ratios only at the end of each reporting period. This lower burden of balance sheet costs enables foreign banks to exploit arbitrage opportunities in the fed funds market, while domestic banks remain largely inactive (Kreicher, McCauley, and McGuire, 2014; Afonso, Entz, and LeSueur, 2013; Ennis and Wolman, 2015; Banegas and Tase, 2022).

2.2 Government-sponsored enterprises

Under the floor system, government-sponsored enterprises (GSEs) have become the primary lenders in the fed funds market. Since 2016, more than 90 percent of total lending has been provided by

GSEs.¹² Among them, the Federal Home Loan Banks (FHLBs) are the most significant lenders, while other GSEs—such as Fannie Mae and Freddie Mac—have been largely inactive (Afonso, Cisternas, Gowen, Miu, and Younger, 2023a).

The FHLBs are financial intermediaries that borrow in global markets and provide whole-sale funding to their members—banks and other eligible institutions—through collateralized loans (known as “advances”). In 2017, their total debt outstanding was about one trillion dollars, and about half of it was held by money market mutual funds. Thus, part of their business is to serve as intermediaries between their members—such as banks—and money market mutual funds (Gissler and Narajabad, 2017).

Unlike depository institutions, FHLBs are not eligible to earn interest on their reserve balances at the Fed. Because reserves are effectively zero-interest assets for them, FHLBs have an incentive to lend even when the fed funds rate is below the IORB rate. Another motive for lending in the fed funds market is to satisfy regulatory liquidity requirements. Consistent with this, they held about 7.5 percent of their assets in fed funds in 2017 (Gissler and Narajabad, 2017).

3 The Model

The model is based on the framework of Lagos and Wright (2005) and Rocheteau and Wright (2005), which generates a role for a medium of exchange. We extend this framework to a two-country setting, following Gomis-Porqueras, Kam, and Lee (2013), Gomis-Porqueras, Kam, and Waller (2017), and Kim (2023), and incorporate multiple banking sectors in the spirit of Williamson (2019, 2023).

There are two countries, *Home* and *Foreign*, representing the United States and the rest of the world, respectively. Each country is populated by three types of domestic agents: *buyers*, *sellers*, and *retail banks*. In addition, there are *unconventional banks* that operate globally by providing financial services in both countries. While retail banks represent traditional depository institutions in each country, unconventional banks represent non-depository financial institutions, including GSEs and money market funds.¹³ For each type, there is a continuum of agents with unit mass.

Parameters and variables associated with the Foreign country are denoted with an asterisk, and their Home counterparts are denoted without an asterisk. When describing the model environment, we focus on the Home country, bearing in mind that there is a Foreign counterpart.

Governments and Fundamental Assets—In each country, the government—consisting of a fiscal authority and a central bank—issues three types of assets: currency, reserves, and government

¹²Figure 10 in Appendix A.1 shows the changes in the shares of fed funds lending by institution type during 2016-2024.

¹³One could introduce GSEs as a separate class of financial institutions that borrow from global money market funds in the repo market and lend to banks in the fed funds market. Although this alternative setup would provide a more direct link between the model and the actual fed funds market, the main results would remain unchanged, since our focus is on replicating arbitrage activity in short-term funding markets arising from the spread between the IORB rate and other short-term interest rates (e.g., the fed funds rate and repo rates). The welfare implications would also remain unchanged, given that global money market funds provide financial services to both domestic and foreign clients.

bonds. Fiscal authorities issue government bonds, denominated in their local currency, and central banks issue currency and reserves through open market operations. Reserves are account balances that retail banks hold at their central banks. Currency and reserves issued by the Home country's central bank are referred to as *Home currency* and *Home reserves*, respectively, and their Foreign counterparts are referred to as *Foreign currency* and *Foreign reserves*.

Timing of Events—Each period is divided into two sequential subperiods: the centralized market (CM) followed by the decentralized market (DM). The CM provides an international platform where all agents from both countries can interact. At the beginning of each CM, all outstanding debts are settled before new debts are issued. Home and Foreign agents then produce goods, trade goods for assets, and consume goods in a perfectly competitive market. During this stage, retail and unconventional banks receive deposits (or equivalently, issue deposit liabilities) and trade interbank loans. After the CM closes, buyers can no longer trade goods or assets but have an opportunity to visit their banks. Whether they choose to do so depends on the type of seller they will meet in the subsequent DM, as specified later.

In the CM, the price of Home (Foreign) currency in units of goods is denoted by ϕ (ϕ^*). Each unit of Home (Foreign) reserves sells at price z_m (z_m^*) and is a claim to one unit of reserves in the subsequent CM. Each Home currency-denominated bond (a *Home bond*, hereafter) sells at price z_b and pays off one unit of Home currency in the following CM. Similarly, each Foreign currency-denominated bond (a *Foreign bond*) sells at price z_b^* and pays off one unit of Foreign currency in the following CM.¹⁴

In contrast to the CM, the DM is a domestic platform in which each buyer is randomly matched with a seller within the country. The terms of trade in the DM are determined through bargaining. For convenience, we assume that the buyer makes a take-it-or-leave-it offer to the seller. The timing of events is summarized in [Figure 4](#).

Preferences—In the CM, buyers can produce goods but cannot consume, whereas sellers can consume goods but cannot produce. Specifically, each buyer incurs disutility H when producing H units of *CM goods*, whereas each seller receives utility X from consuming X units of CM goods. Retail and unconventional banks are active only in the CM, where they can both produce and consume goods. Their per-period preferences are given by $X^i - H^i$, where X^i and H^i denote their consumption and production, respectively, in the CM.¹⁵

In the DM, sellers can produce goods but cannot consume, whereas buyers can consume goods but cannot produce. Each buyer receives utility $u(x)$ from consuming x units of *DM goods*, whereas each seller incurs disutility h from producing h units of DM goods. The utility function $u(\cdot)$ is strictly increasing, strictly concave, and twice continuously differentiable, with properties $u'(0) = \infty$,

¹⁴One could also introduce private assets, such as mortgages, into the model. However, this would not alter the main results, since our focus is on potential arbitrage in the interbank market and its implications for monetary policy.

¹⁵Modeling banks as agents with preferences is not uncommon in the literature. For example, [Diamond \(1984\)](#), [Williamson \(1986\)](#), [Cavalcanti and Wallace \(1999\)](#), and [Gu, Mattesini, Monnet, and Wright \(2013\)](#) consider settings in which some agents endogenously emerge as banks due to their comparative advantage in providing banking services.

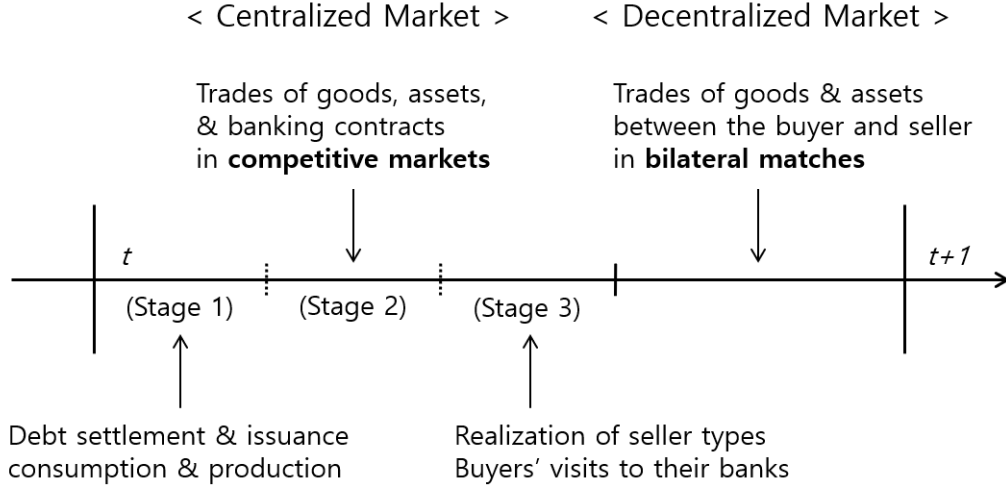


Figure 4: Timing of events

$u'(\infty) = 0$, and $-\frac{xu''(x)}{u'(x)} < 1$ for $x > 0$. Let $x^e \in (0, \infty)$ denote the first-best quantity that solves $u'(x^e) = 1$.

Buyer and Seller Types—There is no record-keeping device (no memory), and agents cannot be forced to work (limited commitment). Thus, unsecured credit arrangements are infeasible, and some assets must be exchanged on the spot or posted as collateral for transactions in the DM. The range of acceptable means of payment in DM transactions, however, varies across seller types.

Depending on the seller's ability to evaluate assets, there are four types of sellers in each country. *Type c* sellers recognize only paper currency issued by the domestic central bank and thus accept only domestic currency. *Type b* sellers accept currency and government bonds issued in the Home country, while *type b** sellers accept those issued in the Foreign country. Finally, *type d* sellers can evaluate all types of assets, including currencies, government bonds, reserves, and any bank liabilities backed by these assets. While the DM trades with *type c* and *type d* sellers capture typical payments using cash and bank deposits, trades with *type b* and *type b** sellers capture wholesale payments that involve government bonds (e.g., repurchase agreements that typically use government bonds as collateral).

In each country, buyers are *ex ante* divided into two groups: retail bank depositors and unconventional bank depositors. Specifically, a fraction α of buyers in the Home country (*Home buyers*, hereafter) are retail bank depositors, while the remaining fraction $1 - \alpha$ are unconventional bank depositors. Similarly, a fraction α^* of buyers in the Foreign country (*Foreign buyers*) are retail bank depositors, while a fraction $1 - \alpha^*$ are unconventional bank depositors.

All buyers face uncertainty about the type of seller they are matched with in the DM, but the set of sellers they may encounter differs by buyer type. Each retail bank depositor meets either a *type c* seller with probability ρ or a *type d* seller with probability $1 - \rho$. In contrast, each unconventional

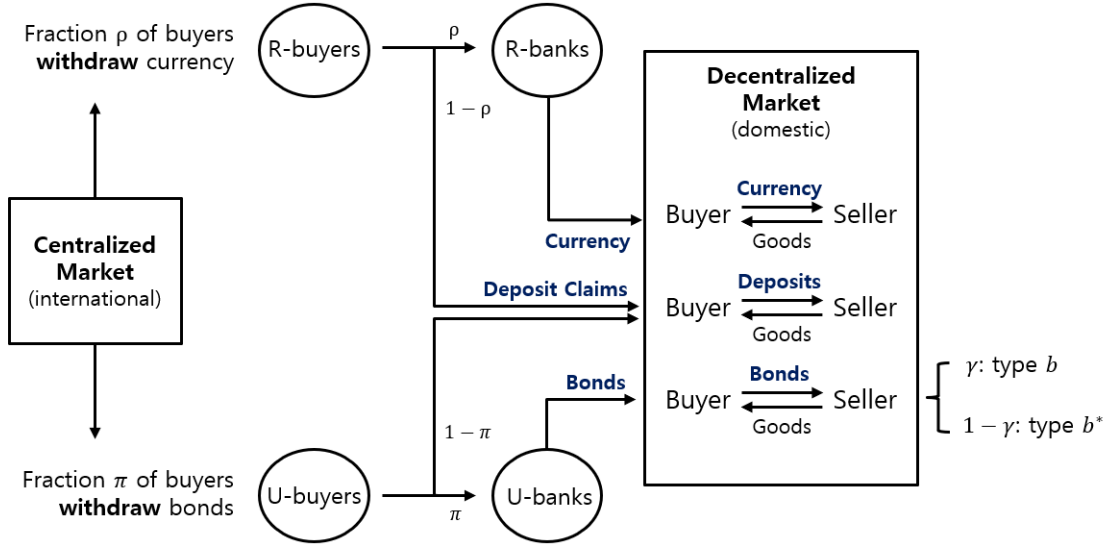


Figure 5: Equilibrium events in the Home country after the centralized market closes

Note: Retail banks and their depositors are denoted by R-banks and R-buyers, respectively. Similarly, unconventional banks and their depositors are denoted by U-banks and U-buyers.

bank depositor meets a type b seller with probability $\pi\gamma$, a type b^* seller with probability $\pi(1 - \gamma)$, or a type d seller with probability $1 - \pi$. Thus, in each country, five distinct types of DM matches arise *ex post*.

Banks and Their Liabilities—When buyers acquire assets in the CM, they do not yet know the type of seller they will meet in the subsequent DM. Buyers learn the seller's type and the set of acceptable means of payment only after the CM closes, but they are able to contact their bank. This means-of-payment shock, together with spatial separation, gives banks a role in efficiently allocating resources across DM meetings, in the spirit of [Diamond and Dybvig \(1983\)](#).

Banks can issue deposit liabilities that are backed by fundamental assets—currency, reserves, and government bonds—but that have a payoff structure distinct from these assets. In particular, banks can design deposit contracts that give depositors the option to withdraw currency or bonds after the CM closes. If a depositor exercises this option, the depositor no longer holds a deposit claim; if the depositor does not exercise the option, the depositor retains a deposit claim that yields a payoff in the next CM.

These deposit contracts provide insurance to depositors against uncertainty regarding their means-of-payment shocks. For example, [Figure 5](#) illustrates equilibrium events in the Home country after the CM. A fraction ρ of retail bank depositors learn that the seller they will meet is type c and visit the bank to withdraw currency. By contrast, a fraction $1 - \rho$ learn that the seller is type d and choose not to visit their bank, because they prefer using deposit claims as a means of payment in the DM rather than currency.

In addition to deposit liabilities, which are specified in the next section, banks can borrow from other banks by issuing interbank loans. Each interbank loan sells at price z_e in the CM and represents a promise to pay one unit of Home currency in the following CM.

Institutional Differences Among Banks—We introduce two key assumptions that reflect observed features of financial regulation. First, retail banks can hold reserve accounts with central banks and earn interest, regardless of their nationality, whereas unconventional banks cannot.¹⁶ Because of this restriction, unconventional banks may have an incentive to lend to retail banks in the interbank market in order to obtain indirect access to reserves.

Second, retail banks in the Home country are subject to a capital requirement that is tighter than the balance sheet constraint faced by their Foreign counterparts. Specifically, retail banks in the Foreign country (as well as unconventional banks) face only a collateral constraint: the total value of their liabilities cannot exceed the total value of their assets. By contrast, the capital requirement implies that the total value of a Home retail bank’s liabilities cannot exceed a fraction $1 - \delta$ of the total value of its assets, where δ denotes the capital ratio. As a result, expanding the size of their balance sheets is more costly for Home retail banks.

These two assumptions capture, in a parsimonious way, the key frictions that have shaped the current landscape of the fed funds market discussed in the previous section. Within this simplified framework, we analyze the conditions under which the model-generated interbank market resembles the observed fed funds market, in which foreign banks borrow from GSEs to engage in arbitrage. We then study the effects of various monetary policies on asset prices and welfare.

3.1 Government budget constraints and policies

We confine our attention to a stationary equilibrium in which all real variables are constant over time and the gross inflation rates are μ and μ^* in the Home and Foreign countries, respectively. This implies that $\frac{\phi_{t+1}}{\phi_t} = \frac{1}{\mu}$ and $\frac{\phi_{t+1}^*}{\phi_t^*} = \frac{1}{\mu^*}$ for all $t \geq 0$. In each country, there is no public debt outstanding at the beginning of period $t = 0$, prior to the issuance of new government liabilities. The fiscal authority issues government bonds denominated in domestic currency and uses the proceeds to make lump-sum transfers of equal size to domestic buyers. The central bank issues its liabilities—domestic currency and reserves—through open market purchases of government bonds.

The consolidated government budget constraints for the two countries in $t = 0$ are given by:

$$\begin{aligned}\bar{c} + z_m \bar{m} + z_b \bar{b} &= \tau_0, \\ \bar{c}^* + z_m^* \bar{m}^* + z_b^* \bar{b}^* &= \tau_0^*,\end{aligned}$$

where $\bar{c}$, $\bar{m}$, and $\bar{b}$ denote the quantities of Home currency, Home reserves, and Home bonds outstanding, in real terms, at the end of period 0 (and in each subsequent period). Similarly, $\bar{c}^*$,

¹⁶There may be institutional differences across countries. For example, not all foreign banks are eligible to hold reserve accounts with the Bank of Canada. However, in our model, prohibiting Home retail banks from holding reserves at the Foreign central bank does not affect the main results of the paper, as we show later.

<Home Central Bank>		<Foreign Central Bank>	
Assets	Liabilities	Assets	Liabilities
Home bonds $v - z_b \bar{b}$	Home currency $\bar{c}$	Foreign bonds $v^* - z_b^* \bar{b}^*$	Foreign currency $\bar{c}^*$
	Home reserves $z_m \bar{m}$		Foreign reserves $z_m^* \bar{m}^*$

Table 1: Balance sheets of central banks

$\bar{m}^*$, and $\bar{b}^*$ denote their Foreign counterparts.¹⁷ Finally, τ_0 and τ_0^* denote the real quantities of lump-sum transfers made by the Home and Foreign fiscal authorities, respectively, in period 0.

In each subsequent period, fiscal authorities collect taxes to repay outstanding debts before issuing new liabilities. Central banks receive payoffs from their asset holdings, redeem their debts, and transfer any profits to their domestic fiscal authorities. The consolidated government budget constraints for $t \geq 1$ are:

$$\begin{aligned}\bar{c} + z_m \bar{m} + z_b \bar{b} &= \frac{\bar{c} + \bar{m} + \bar{b}}{\mu} + \tau, \\ \bar{c}^* + z_m^* \bar{m}^* + z_b^* \bar{b}^* &= \frac{\bar{c}^* + \bar{m}^* + \bar{b}^*}{\mu^*} + \tau^*,\end{aligned}$$

where τ and τ^* denote the real quantities of net lump-sum transfers to Home and Foreign buyers, respectively, for period $t \geq 1$. In each of the above equations, the left-hand side represents the total value of the consolidated government debts outstanding, while the right-hand side represents the redemption value of debts plus the net transfers to buyers.

Following [Williamson \(2019\)](#) and [Kim \(2023\)](#), we assume that the Home and Foreign fiscal authorities comply with the following fiscal policy rules:

$$v = \bar{c} + z_m \bar{m} + z_b \bar{b}, \tag{1}$$

$$v^* = \bar{c}^* + z_m^* \bar{m}^* + z_b^* \bar{b}^*. \tag{2}$$

Here, v and v^* , which represent the real values of consolidated government liabilities, are exogenously given. Each fiscal authority adjusts the quantities of lump-sum transfers to target v and v^* for every $t \geq 0$. Given these fiscal rules, the balance sheets of central banks are displayed in [Table 1](#).

In practice, central banks typically adopt one of two monetary policy operating frameworks: a *corridor system* or a *floor system*. Under a corridor system, the central bank targets a short-term interest rate—typically an overnight rate—and actively intervenes in the market to maintain it. The policy rate fluctuates within a corridor bounded below by the interest rate on reserves and above by the central bank lending rate. By contrast, under a floor system, the central bank supplies a large quantity of reserves, and the interest rate on reserves directly influences the short-term interest rate,

¹⁷For simplicity, we assume that central banks do not hold foreign currency or foreign currency-denominated assets. This assumption is innocuous, as the main findings of the paper would remain unchanged without it. However, relaxing this assumption would allow us to analyze the effects of central bank interventions in the foreign exchange market (see Appendix A.2 for details).

rather than merely acting as a lower bound.

Our goal is to replicate the landscape of the U.S. overnight lending market since late 2008. Specifically, U.S. banks have been largely inactive, while U.S. branches and agencies of foreign banks have actively engaged in arbitrage by borrowing funds at a lower rate and depositing them in reserve accounts to earn a higher interest. For this purpose, we assume that both Home and Foreign central banks operate monetary policy under a floor system.¹⁸ That is, the Home central bank sets the price of Home reserves z_m (or equivalently, the nominal interest rate on Home reserves $R_m = \frac{1}{z_m} - 1$) and the size of its balance sheet, denoted by $\omega \equiv \bar{c} + z_m \bar{m}$. Similarly, the Foreign central bank determines the price of Foreign reserves z_m^* (or $R_m^* = \frac{1}{z_m^*} - 1$) and the size of its balance sheet $\omega^* \equiv \bar{c}^* + z_m^* \bar{m}^*$.

4 Characterization of Equilibrium

4.1 Retail banks and unconventional banks

Buyers write deposit contracts with banks in the CM before learning which means of payment (equivalently, which seller type) will be acceptable in the subsequent DM. In equilibrium, retail banks in the Home country (*Home banks*, hereafter) and those in the Foreign country (*Foreign banks*) offer deposit contracts to their domestic retail bank depositors, while unconventional banks issue deposit contracts to their depositors in both countries. As in Williamson (2019) and Kim (2023), these deposit contracts grant each depositor the option to withdraw a means of payment after learning the type of seller they will encounter.

Consider a Home bank offering deposit contracts to retail bank depositors in the Home country (*Home bank depositors*, hereafter). Because Home bank depositors meet a type c seller—who accepts only domestic currency—with positive probability, Home banks will endogenously offer a contract that allows depositors to withdraw Home currency once the CM closes. If a depositor meets a type d seller—who accepts all types of assets—they will not withdraw currency and will use their deposit claims as the means of payment in the DM.

Let (k_h, c_h, d_h) denote the terms of a deposit contract offered by the Home bank, where k_h is the quantity of goods deposited in the CM, c_h is the quantity of Home currency the depositor may withdraw after the CM, and d_h is the quantity of claims to goods in the next CM for those who do not withdraw currency. Let $(m_h, m_h^*, b_h, b_h^*, e_h)$ denote the Home bank's asset portfolio, where m_h and m_h^* are its holdings of Home and Foreign reserves, b_h and b_h^* are its holdings of Home and Foreign bonds, and e_h is its interbank borrowing.¹⁹ All quantities are expressed in units of current

¹⁸This assumption is justified, as major central banks—including the European Central Bank, the Bank of England, and the Bank of Japan—have operated under a floor system. Alternatively, one could assume that the Home central bank uses a floor system while the Foreign central bank operates under a corridor system. This can be modeled by setting $z_m^* = 1$, so that no banks wish to hold reserves at the Foreign central bank. However, the main results of this paper would remain unchanged.

¹⁹Because the Home bank weakly prefers interest-bearing Foreign bonds or Foreign reserves to zero-interest Foreign currency, it does not hold Foreign currency in equilibrium.

CM goods, except for d_h , which is expressed in units of next-period CM goods.

Then, the Home bank's problem in equilibrium can be written as:

$$\max_{k_h, c_h, d_h, m_h, m_h^*, b_h, b_h^*, e_h} \left[-k_h + \rho u \left(\frac{\beta c_h}{\mu} \right) + (1 - \rho) u(\beta d_h) \right] \quad (3)$$

subject to

$$k_h + z_e e_h - \left[\begin{array}{c} \rho c_h + z_m m_h + z_m^* m_h^* \\ + z_b b_h + z_b^* b_h^* \end{array} \right] + \beta \left[-(1 - \rho) d_h + \frac{m_h + b_h - e_h}{\mu} + \frac{m_h^* + b_h^*}{\mu^*} \right] \geq 0, \quad (4)$$

$$-(1 - \rho) d_h - \frac{e_h I_h}{\mu} - \frac{\delta \rho c_h}{\mu} + \frac{(1 - \delta)[m_h + b_h - e_h(1 - I_h)]}{\mu} + \frac{(1 - \theta)(1 - \delta)(b_h^* + m_h^*)}{\mu^*} \geq 0, \quad (5)$$

$$k_h, c_h, d_h, m_h, m_h^*, b_h, b_h^* \geq 0. \quad (6)$$

As expressed in the objective function (3), the Home bank chooses a contract (k_h, c_h, d_h) that maximizes the representative Home bank depositor's expected utility. In each DM meeting, the buyer makes a take-it-or-leave-it offer to the seller. This implies that in a meeting with a type c seller, the Home bank depositor trades c_h units of Home currency for $\frac{\beta c_h}{\mu}$ units of DM goods, while in a meeting with a type d seller, they exchange d_h units of claims to deposits for βd_h units of goods.

Inequality (4) states that the Home bank's discounted net payoff must be nonnegative. In the CM, the bank receives k_h units of goods from depositors and acquires an asset portfolio consisting of Home currency ρc_h , Home reserves m_h , Foreign reserves m_h^* , Home bonds b_h , Foreign bonds b_h^* , and interbank borrowing e_h . If the bank borrows in the interbank market, then $e_h > 0$ and $I_h = 1$; if it lends, then $e_h < 0$ and $I_h = 0$. After the CM closes, a fraction ρ of depositors learn that they will meet a type c seller in the DM and therefore withdraw c_h units of Home currency. The remaining fraction $1 - \rho$ do not withdraw currency, and in the next CM the bank pays each deposit holder d_h units of goods while collecting the payoffs from its assets.

A deposit contract is essentially a debt contract through which the bank borrows from depositors. Because of limited commitment and no memory, the bank's liabilities must be secured by its assets, generating a collateral constraint. However, unlike other banks, Home banks face an additional capital requirement imposed by the Home government, represented by inequality (5). This capital constraint ensures that the value of the bank's liabilities cannot exceed a fraction $1 - \delta$ of the risk-adjusted value of its assets at maturity. Moreover, only a fraction $1 - \theta$ of foreign currency-denominated assets is counted toward the risk-adjusted asset value.²⁰

A Foreign bank's problem in equilibrium is analogous to that of the Home bank, except for the collateral constraint it faces. A Foreign bank writes deposit contracts with retail bank depositors in

²⁰In practice, all else equal, foreign currency-denominated assets are considered riskier than domestic currency-denominated assets because their values fluctuate with exchange rate movements. As a result, foreign currency-denominated assets typically receive larger haircuts when pledged as collateral. See Kim (2023) who studies the implications of such haircuts for foreign exchange interventions.

the Foreign country (*Foreign bank depositors*, hereafter), who meet either a type c or type d seller in the following DM. Let (k_f, c_f, d_f) denote the terms of the Foreign bank's deposit contracts, where k_f is the quantity of goods deposited in the CM, c_f is the quantity of Foreign currency the depositor may withdraw after the CM closes, and d_f is the quantity of claims to goods in the following CM for those who do not withdraw currency. Let $(m_f, m_f^*, b_f, b_f^*, e_f)$ denote the Foreign bank's asset portfolio, where m_f and m_f^* are its holdings of Home and Foreign reserves, b_f and b_f^* are its holdings of Home and Foreign bonds, and e_f is the quantity of interbank borrowing.

Then, the Foreign bank's problem in equilibrium is given by:

$$\max_{k_f, c_f^*, d_f, m_f, m_f^*, b_f, b_f^*, e_f} \left[-k_f + \rho u \left(\frac{\beta c_f^*}{\mu^*} \right) + (1 - \rho) u(\beta d_f) \right] \quad (7)$$

subject to

$$k_f + z_e e_f - \left[\rho c_f^* + z_m m_f + z_m^* m_f^* + z_b b_f + z_b^* b_f^* \right] + \beta \left[-(1 - \rho) d_f + \frac{m_f + b_f - e_f}{\mu} + \frac{b_f^* + m_f^*}{\mu^*} \right] \geq 0, \quad (8)$$

$$-(1 - \rho) d_f - \frac{e_f I_f}{\mu} + \frac{(1 - \theta)[m_f + b_f - e_f(1 - I_f)]}{\mu} + \frac{b_f^* + m_f^*}{\mu^*} \geq 0, \quad (9)$$

$$k_f, c_f^*, d_f, m_f, m_f^*, b_f, b_f^* \geq 0. \quad (10)$$

The Foreign bank is subject to a collateral constraint, instead of a capital constraint, implying that it is less restricted by the size of its balance sheet than the Home bank.²¹ The collateral constraint, inequality (9), ensures that the Foreign bank's net payoff from repaying its debts is at least as large as its net payoff from defaulting.

Unconventional banks write deposit contracts with buyers in both Home and Foreign countries. With probability $\pi\gamma$, an unconventional bank depositor meets a type b seller who accepts only Home bonds in the DM, and with probability $\pi(1 - \gamma)$, they meet a type b^* seller who accepts only Foreign bonds. With the remaining probability $1 - \pi$, the depositor meets a type d seller, in which case they can use their claims on deposits as a medium of exchange. Considering these liquidity needs, unconventional banks design contracts that allow depositors to withdraw either Home bonds or Foreign bonds after the CM closes.

Let $(k_u, \hat{b}_u, \hat{b}_u^*, d_u)$ denote the terms of an unconventional bank's deposit contract, where k_u is the quantity of goods deposited in the CM, $\hat{b}_u$ and $\hat{b}_u^*$ are the quantities of Home and Foreign bonds the depositor may withdraw after the CM closes, and d_u is the quantity of claims to goods in the next CM for those who do not withdraw bonds. Let (b_u, b_u^*, e_u) denote the unconventional bank's asset portfolio, where b_u and b_u^* are its holdings of Home and Foreign bonds, and e_u is the bank's net interbank borrowing.

Then, the unconventional bank's problem in equilibrium can be written as:

²¹The collateral constraint can be interpreted as the capital constraint with $\delta = 0$.

$$\max_{k_u, \hat{b}_u, \hat{b}_u^*, d_u, b_u, b_u^*, e_u} \left[-k_u + \pi\gamma u \left(\frac{\beta \hat{b}_u}{\mu} \right) + \pi(1-\gamma)u \left(\frac{\beta \hat{b}_u^*}{\mu^*} \right) + (1-\pi)u(\beta d_u) \right] \quad (11)$$

subject to

$$k_u + z_e e_u - z_b b_u - z_b^* b_u^* + \beta \left[-(1-\pi)d_u + \frac{b_u - \pi\gamma \hat{b}_u - e_u}{\mu} + \frac{b_u^* - \pi(1-\gamma)\hat{b}_u^*}{\mu^*} \right] \geq 0, \quad (12)$$

$$-(1-\pi)d_u - \frac{e_u}{\mu} + \frac{b_u - \pi\gamma \hat{b}_u}{\mu} + \frac{(1-\theta)[b_u^* - \pi(1-\gamma)\hat{b}_u^*]}{\mu^*} \geq 0, \quad (13)$$

$$k_u, d_u, b_u, b_u^* \geq 0; \quad 0 \leq \hat{b}_u \leq b_u; \quad 0 \leq \hat{b}_u^* \leq b_u^*. \quad (14)$$

Similar to the Foreign bank's problem, the unconventional bank is subject to a collateral constraint. We assume that unconventional banks are based in the Home country and that their balance sheets are denominated in the Home currency, so only a fraction $1 - \theta$ of assets denominated in Foreign currency counts toward the value of collateral.

In the model, Home banks face a higher balance sheet cost than Foreign banks and unconventional banks because they are subject to tighter balance sheet constraints. This assumption reflects the empirical observation that traditional banks—such as U.S. depository institutions—must comply with tighter financial regulations than other financial institutions. It also captures differences in leverage ratio calculations across jurisdictions, which make U.S. banks face a tighter balance sheet constraint than do the U.S. branches and agencies of foreign banks.

4.2 Definition of equilibrium

We define and characterize a stationary equilibrium in which collateral constraints matter for the behavior of banks. In this model, collateral constraints bind when collateral is scarce, and the supply of collateralizable assets is partly determined by fiscal policies, v and v^* . For our purposes, we focus on cases where v and v^* are sufficiently low such that collateral constraints bind in equilibrium.

In equilibrium, each bank's discounted net payoff must be zero. Otherwise, a bank would have an incentive to adjust its deposit contract to attract more depositors and earn a higher payoff, which would contradict equilibrium. Thus, constraints (4), (8), and (12) must hold with equality in equilibrium.

Let x_c and x_d denote the consumption quantities of Home bank depositors in DM meetings with a type c seller and a type d seller, respectively. That is, $x_c \equiv \frac{\beta c_h}{\mu}$ and $x_d \equiv \beta d_h$. The first-order conditions for the Home bank's problem can then be written as:

$$\mu = \beta[u'(x_c) - \delta u'(x_d) + \delta], \quad (15)$$

$$z_b \geq \frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu}, \quad (16)$$

$$z_b^* \geq \frac{\beta\{(1-\delta)[(1-\theta)u'(x_d) + \theta] + \delta\}}{\mu^*}, \quad (17)$$

$$z_m = \frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu}, \quad (18)$$

$$z_m^* \geq \frac{\beta\{(1-\delta)[(1-\theta)u'(x_d) + \theta] + \delta\}}{\mu^*}, \quad (19)$$

$$z_e \leq \frac{\beta u'(x_d)}{\mu}, \quad (20)$$

$$z_e \geq \frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu}. \quad (21)$$

Equation (15) implies that, in equilibrium, the Home bank always holds a positive quantity of Home currency to meet the demand for currency withdrawals. The bank may or may not hold government bonds and reserves. Each of equations (16)-(19) holds with equality if the bank holds a positive quantity of the corresponding asset—Home bonds, Foreign bonds, Home reserves, and Foreign reserves, respectively. This ensures that the discounted net payoff from holding the asset is zero in equilibrium. If the bank chooses not to hold an asset, the corresponding condition holds with inequality; that is, the bank avoids holding the asset if its discounted net payoff would be negative. The bank may or may not trade in the interbank market. If it borrows in the market, equation (20) holds with equality and $e_h > 0$. If it lends, (21) holds with equality and $e_h < 0$.

Similarly, we can derive the first-order conditions for the Foreign bank's problem. Let x_c^* and x_d^* denote the consumption quantities of Foreign bank depositors in DM meetings with a type c seller and a type d seller, respectively. That is, $x_c^* \equiv \frac{\beta c_f^*}{\mu^*}$ and $x_d^* \equiv \beta d_f$. Then, the following conditions must hold in equilibrium:

$$\mu^* = \beta u'(x_c^*), \quad (22)$$

$$z_b \geq \frac{\beta[(1-\theta)u'(x_d^*) + \theta]}{\mu}, \quad (23)$$

$$z_b^* \geq \frac{\beta u'(x_d^*)}{\mu^*}, \quad (24)$$

$$z_m \geq \frac{\beta[(1-\theta)u'(x_d^*) + \theta]}{\mu}, \quad (25)$$

$$z_m^* = \frac{\beta u'(x_d^*)}{\mu^*}, \quad (26)$$

$$z_e \leq \frac{\beta u'(x_d^*)}{\mu}, \quad (27)$$

$$z_e \geq \frac{\beta[(1-\theta)u'(x_d^*) + \theta]}{\mu}. \quad (28)$$

Next, let x_b , x_b^* , and x_d^u denote the consumption quantities of unconventional bank depositors in DM meetings with a type b seller, a type b^* seller, and a type d seller, respectively. This implies that $x_b \equiv \frac{\beta \hat{b}_u}{\mu}$, $x_b^* \equiv \frac{\beta \hat{b}_u^*}{\mu^*}$, and $x_d^u \equiv \beta d_u$. Then, the first-order conditions for the unconventional

bank's problem can be written as:

$$z_b = \frac{\beta u'(x_b)}{\mu}, \quad (29)$$

$$u'(x_b) \geq u'(x_d^u), \quad \text{if } \pi\gamma x_b = \frac{\beta b_u}{\mu} \quad (30)$$

$$u'(x_b) = u'(x_d^u), \quad \text{if } \pi\gamma x_b < \frac{\beta b_u}{\mu} \quad (31)$$

$$z_b^* = \frac{\beta u'(x_b^*)}{\mu^*}, \quad (32)$$

$$u'(x_b^*) \geq (1 - \theta)u'(x_d^u) + \theta, \quad \text{if } \pi(1 - \gamma)x_b^* = \frac{\beta b_u^*}{\mu^*} \quad (33)$$

$$u'(x_b^*) = (1 - \theta)u'(x_d^u) + \theta, \quad \text{if } \pi(1 - \gamma)x_b^* < \frac{\beta b_u^*}{\mu^*} \quad (34)$$

$$z_e = \frac{\beta u'(x_d^u)}{\mu}. \quad (35)$$

Equations (29)-(34) determine the quantities of bonds that depositors may withdraw after the CM closes, as well as whether the unconventional bank holds bonds in excess of these withdrawal needs. Equation (35) ensures that the unconventional bank is indifferent between borrowing and lending in the interbank market.

Asset market clearing conditions, where the sum of demands equals the corresponding supply, are given by:

$$\alpha \rho c_h = \bar{c}, \quad (36)$$

$$\alpha b_h + \alpha^* b_f + (2 - \alpha - \alpha^*) b_u = \bar{b}, \quad (37)$$

$$\alpha m_h + \alpha^* m_f = \bar{m}, \quad (38)$$

$$\alpha^* \rho c_f^* = \bar{c}^*, \quad (39)$$

$$\alpha b_h^* + \alpha^* b_f^* + (2 - \alpha - \alpha^*) b_u^* = \bar{b}^*, \quad (40)$$

$$\alpha m_h^* + \alpha^* m_f^* = \bar{m}^*, \quad (41)$$

$$\alpha e_h + \alpha^* e_f + (2 - \alpha - \alpha^*) e_u = 0, \quad (42)$$

Equations (36)-(38) and (42) correspond to the Home country's markets for currency, government bonds, reserves, and interbank loans, while equations (39)-(41) correspond to the Foreign country's markets for currency, government bonds, and reserves.

Because the law of one price holds in the CM, the nominal depreciation rate of the Home currency from the current period to the next is characterized by:

$$\frac{\xi_{+1}}{\xi} = \frac{\mu}{\mu^*}. \quad (43)$$

That is, the ratio of the Home inflation rate to the Foreign inflation rate determines the currency

depreciation rate in equilibrium. Using (29) and (32), this depreciation rate is also expressed as:

$$\frac{\xi_{+1}}{\xi} = \frac{z_b^*}{z_b} \cdot \frac{u'(x_b)}{u'(x_b^*)}. \quad (44)$$

Equation (44) shows that the nominal interest rate differential, z_b^*/z_b , is not the sole determinant of the depreciation rate. A real interest rate differential, represented by $u'(x_b)/u'(x_b^*)$, can emerge, potentially leading to a breakdown of uncovered interest rate parity. This occurs because government bonds serve not only as a store of value but also as a means of payment and collateral. Their non-pecuniary payoff generates a liquidity premium, lowering the real interest rate. Moreover, differences in liquidity needs across countries contribute to variation in real interest rates for government bonds.

Now, we can define an equilibrium as follows.

Definition Given exogenous fiscal policies (v, v^*) and monetary policies $(z_m, \omega, z_m^*, \omega^*)$, a stationary equilibrium consists of consumption quantities in DM meetings $(x_c, x_d, x_c^*, x_d^*, x_b, x_b^*, x_d^u)$, asset holdings of banks $(\{c_i, c_i^*, m_i, m_i^*, b_i, b_i^*, e_i\}_{i=h,f,u})$, asset prices (z_b, z_b^*, z_e) , gross inflation rates (μ, μ^*) , and the nominal depreciation rate of the Home currency $\frac{\xi_{+1}}{\xi}$ that satisfy (5), (9), and (13) (capital and collateral constraints), (1)-(2) (fiscal policy rules), (15)-(35) (first-order conditions for bank problems), (36)-(42) (asset market-clearing conditions), and (43) (exchange rate condition).

Proposition 1 Suppose v and v^* are sufficiently small such that

$$(1 - \theta)v + v^* < x^e \left\{ (1 - \rho) \left[\frac{\alpha(1 - \theta)}{1 - \delta} + \alpha^* \right] + (2 - \alpha - \alpha^*)[1 - \theta(1 - \pi + \pi\gamma)] \right\},$$

where $u'(x^e) = 1$. Then, in equilibrium, constraints (5), (9), and (13) hold with equality, while $b_h^* = m_h^* = 0$ and $b_u^* = \pi(1 - \gamma)\hat{b}_u^*$.

Under the condition presented in Proposition 1, the capital constraint for Home banks and the collateral constraints for Foreign and unconventional banks bind in equilibrium. In addition, Home banks do not hold Foreign bonds or reserves—that is, $b_h^* = m_h^* = 0$ —as they strictly prefer to hold other assets to satisfy the capital requirement. Unconventional banks hold Foreign bonds only to meet depositors' withdrawal needs, implying $b_u^* = \pi(1 - \gamma)\hat{b}_u^*$. In what follows, we restrict attention to cases where v and v^* satisfy this condition.

Given that the capital and collateral constraints bind, the equilibrium conditions allow us to express each buyer's welfare as:

$$\begin{aligned} \mathcal{U}_h &= \rho [u(x_c) - x_c u'(x_c)] + (1 - \rho) [u(x_d) - x_d u'(x_d)], \\ \mathcal{U}_f &= \rho [u(x_c^*) - x_c^* u'(x_c^*)] + (1 - \rho) [u(x_d^*) - x_d^* u'(x_d^*)], \\ \mathcal{U}_u &= \pi\gamma [u(x_b) - x_b u'(x_b)] + \pi(1 - \gamma) [u(x_b^*) - x_b^* u'(x_b^*)] + (1 - \pi) [u(x_d^u) - x_d^u u'(x_d^u)], \end{aligned}$$

where $\mathcal{U}_h$ denotes the welfare of Home bank depositors, $\mathcal{U}_f$ the welfare of Foreign bank depositors, and $\mathcal{U}_u$ the welfare of unconventional bank depositors. Then, the welfare measure for each country is defined as the weighted sum of utilities across domestic agents, given by:

$$\begin{aligned}\mathcal{W} &= \alpha \mathcal{U}_h + (1 - \alpha) \mathcal{U}_u, \\ \mathcal{W}^* &= \alpha^* \mathcal{U}_f + (1 - \alpha^*) \mathcal{U}_u,\end{aligned}$$

where $\mathcal{W}$ represents welfare in the Home country and $\mathcal{W}^*$ represents welfare in the Foreign country.

5 Equilibrium with Inactive Interbank Market: A Baseline Case

In the previous section, we established a sufficient condition under which balance sheet constraints—capital and collateral constraints—bind in equilibrium. In this environment, monetary policies of the two countries affect asset prices by altering the severity of collateral shortages in financial markets. In particular, the size of the Home central bank's balance sheet ω can directly affect the tightness of banks' balance sheet constraints. This is because an increase in ω involves the central bank's swap of reserves for government bonds, thereby changing the composition of assets held by banks.

When ω is sufficiently small, the supply of Home bonds in financial markets is large enough that all types of banks hold these bonds in their portfolios. As ω increases, however, the quantities of reserves and currency issued by the Home central bank grow relative to the stock of Home bonds. Eventually, the price of Home bonds rises above that of reserves, leading retail banks to stop holding Home bonds. When ω becomes very large, unconventional banks hold Home bonds only to satisfy depositors' withdrawal requests, implying that Home bonds no longer serve as collateral.

Proposition 2 If $\delta > \theta$, there exists a set of parameters leading to an active interbank market where Foreign banks borrow from unconventional banks.

The above proposition indicates that a tight capital constraint—captured by a high value of δ —prevents Home banks from participating in the interbank market, whereas Foreign banks may borrow from unconventional banks. As we will show, when ω (the size of the Home central bank's balance sheet) becomes sufficiently large, the quantity of Home bonds circulating in financial markets becomes sufficiently small. In this case, unconventional banks begin to hold interbank loans as an alternative to Home bonds. At the same time, Foreign banks start borrowing in the interbank market at a relatively low interest rate in order to earn a higher return on reserves.

Proposition 3 Suppose that v^* is sufficiently large such that

$$v^* > \omega^* + (2 - \alpha - \alpha^*)(1 - \gamma)\pi x_b^* u'(u_b^*)$$

in equilibrium. Then, $b_f^* > 0$, and financial markets are interconnected in that changes in the Foreign bond price affect other asset prices and vice versa.

<Home banks>		<Foreign banks>		<Unconventional banks>	
Asset	Liability	Asset	Liability	Asset	Liability
Home currency	Home currency	Foreign currency	Foreign currency	Foreign bonds	Foreign bonds
Home Reserves	Deposits	Reserves	Deposits	Home bonds	Home bonds
Home bonds		Bonds			Deposits

Figure 6: Bank asset portfolios in equilibrium (Case 1)

Proposition 3 presents a condition under which financial markets become interconnected across countries. If the real value of consolidated Foreign government liabilities v^* is sufficiently small, Foreign bonds are used only as a means of payment, and the market for Foreign bonds is completely segmented from other asset markets. In this case, the price of Foreign bonds is independent of other asset prices. However, if v^* is sufficiently large, Foreign bonds serve not only as a means of payment but also as collateral. In this situation, changes in the price of Foreign bonds are accompanied by changes in the prices of other collateralizable assets. In what follows, we focus on cases where v^* is large enough to satisfy the condition in Proposition 3.

5.1 Case 1: Home bonds are held by both retail and unconventional banks

Here, we examine an equilibrium in which a sufficiently large supply of Home bonds induces both retail and unconventional banks to hold Home bonds in their portfolios (see Figure 6 for details). If the supply of Home bonds is sufficiently large relative to that of central bank liabilities (i.e., currency and reserves), no interest rate gap arises between Home bonds and reserves. In this situation, retail banks have no incentive to engage in arbitrage, and unconventional banks have no reason to seek alternative interest-bearing assets. Consequently, the interbank market remains inactive.

Suppose that δ , θ , v , and v^* satisfy the conditions in Propositions 1-3. Then, noting that the first-order conditions (16) and (23)-(25) hold with equality, and that $\pi\gamma\hat{b}_u < b_u$ and $\pi(1-\gamma)\hat{b}_u^* = b_u^*$, we can derive the following equilibrium conditions:

$$(1 - \delta)u'(x_d) + \delta = (1 - \theta)u'(x_d^*) + \theta, \quad (45)$$

$$z_m = \frac{(1 - \delta)u'(x_d) + \delta}{u'(x_c) - \delta u'(x_d) + \delta}, \quad (46)$$

$$z_m^* = \frac{u'(x_d^*)}{u'(x_c^*)}, \quad (47)$$

$$x_b^* = x_d^*, \quad (48)$$

$$u'(x_d^u) = u'(x_b) = (1 - \delta)u'(x_d) + \delta. \quad (49)$$

Equations (45) and (49) imply that the consumption quantities in DM meetings with type d sellers are positively correlated across countries. Equations (46) and (47) describe how monetary policies affect the DM consumption quantities in both countries. In particular, raising the target price of reserves (equivalently, lowering their nominal interest rate) increases the consumption quantity in DM meetings involving currency (x_c, x_c^*) relative to other meetings. The resulting structure of asset prices is:

$$\frac{\beta[(1 - \delta)u'(x_d) + \delta]}{\mu} = z_m = z_e = \frac{\beta u'(x_d^u)}{\mu} = \frac{\beta u'(x_b)}{\mu} = z_b, \quad (50)$$

$$\frac{\beta u'(x_d^*)}{\mu^*} = z_m^* = z_b^* = \frac{\beta u'(x_b^*)}{\mu^*}. \quad (51)$$

Since the prices of Home reserves, interbank loans, and Home bonds are identical, retail banks have no incentive to arbitrage by borrowing in the interbank market and depositing reserves at the Home central bank. Unconventional banks are also indifferent between lending in the interbank market and holding Home bonds, as both assets serve as collateral at the margin. Thus, the interbank market remains inactive due to the lack of demand for interbank loans.

Given the condition in Proposition 3 that ensures endogenously interconnected asset markets, all banks are subject to a consolidated balance sheet constraint within the global financial system. From (1), (2), (5), (9), (13), and (36)-(50), noting that $b_h^* = m_h^* = 0$, the consolidated balance sheet constraint can be written as:

$$\begin{aligned} & \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1 - \delta} \right] + \alpha(1 - \rho)x_d \left[u'(x_d) + \frac{\delta}{1 - \delta} \right] \\ & + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1 - \theta)u'(x_d^*)} \right] + \alpha^*(1 - \rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1 - \theta} \right] \\ & + (2 - \alpha - \alpha^*) \left\{ (1 - \pi + \pi\gamma)x_b u'(x_b) + \pi(1 - \gamma)x_b^* \left[u'(x_b^*) + \frac{\theta}{1 - \theta} \right] \right\} \\ & = v + \left[1 + \frac{\theta}{(1 - \theta)u'(x_d^*)} \right] v^*. \end{aligned} \quad (52)$$

Equation (52) states that, in equilibrium, the aggregate demand for collateral (the left-hand side) must equal its aggregate supply (the right-hand side). The quantities of collateral demanded and supplied are adjusted to account for the balance sheet costs faced by banks.

Using equations (45)-(52), we can solve for the consumption quantities in the DM meetings $(x_c, x_d, x_c^*, x_d^*, x_b, x_b^*, x_d^u)$ given monetary and fiscal policies $(z_m, z_m^*, \omega, \omega^*, v, v^*)$. Then, equations (15) and (22) solve for Home and Foreign inflation rates (μ, μ^*) , while equations (50)-(51) determine

the prices of Home bonds, Foreign bonds, and interbank loans (z_b, z_b^*, z_e) . Finally, equation (43) determines the nominal depreciation rate of the Home currency $(\frac{\xi+1}{\xi})$.

For this type of equilibrium to exist, the equilibrium solution must satisfy:

$$0 \leq \omega \leq v - (2 - \alpha - \alpha^*) \{ (1 - \pi) x_d^u u'(x_d^u) + \pi \gamma x_b u'(x_b) \}, \quad (53)$$

According to condition (53), the size of the Home central bank's balance sheet ω must be sufficiently small so that a sufficiently large supply of Home bonds in financial markets induces retail banks to hold both Home reserves and Home bonds.

Discussion—In our model, a sufficiently small balance sheet of the Home central bank leads to no interbank transactions. This occurs because all interest rates—the rate on reserves, the rate on interbank loans, and the rate on government bonds—are identical in equilibrium, so banks have no incentive to borrow or lend in the interbank market. In practice, retail banks also use reserves as a means of payment to settle interbank transactions, and scarce reserves can affect the interbank lending rate, as was the case before 2008. For example, [Armenter and Lester \(2017\)](#) and [Williamson \(2023\)](#) incorporate idiosyncratic shocks in their models to highlight the role of reserves as a means of payment and examine the implications of the Fed's balance sheet contraction. For our purposes, we abstract from this feature, as the interbank lending rate since 2008 has been largely determined by arbitrage transactions.²²

Now, we analyze the effects of the Home central bank's monetary policy on the Home country and its transmission to the Foreign country. Since changes in the size of the central bank's balance sheet ω are irrelevant in this case, we focus on the effects of a conventional interest rate policy.²³

5.1.1 Conventional easing: a fall in the nominal interest rate on reserves

Suppose that the Home central bank decreases the nominal interest rate on Home reserves (or equivalently, increases z_m , since $R_m = \frac{1}{z_m} - 1$). From (46), an increase in z_m induces an increase in x_c and a decrease in x_d in equilibrium. This occurs because Home currency is substituted for Home reserves, which increases Home buyers' consumption in DM meetings with type c sellers and decreases their consumption in meetings with type d sellers. From (45), (48), and (49), a decrease in x_d is accompanied by decreases in x_d^* , x_d^u , x_b , and x_b^* . This comovement arises because a higher price of Home reserves raises the demand for government bonds, driving up their prices and lowering their real interest rates. The resulting reduction in the redemption value of these assets effectively reduces the stock of collateral in the financial system. As a result, the consumption quantities in DM meetings involving government bonds (x_b, x_b^*) and in meetings supported by collateral-backed deposits (x_d, x_d^*, x_d^u) all decrease. The reduction in the stock of collateral also leads to a substitution

²²If the settlement of interbank transactions is relevant, the fed funds rate would exceed the interest rate on reserves, which occurred only in 2019 over the past decade (see [Figure 1](#)).

²³Throughout the paper, we focus on the effects of conventional and unconventional monetary easing. The effects of monetary tightening are symmetric to those of easing.

x_c	x_d	x_c^*	x_d^*	x_b	x_b^*	x_d^u	μ	μ^*	$\frac{\xi+1}{\xi}$
$\uparrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\uparrow$	$\downarrow$
z_m	z_e	z_b	$z_e - z_m$		$z_b - z_e$		z_b^*	$\mathcal{W}$	$\mathcal{W}^*$
$\uparrow$	$\uparrow$	$\uparrow$	$\cdot$		$\cdot$		$\cdot$	$?$	$\downarrow$

Table 2: Effects of an increase in z_m (Case 1)

of Foreign reserves for Foreign currency, as Foreign reserves become more valuable as collateral. This, in turn, reduces x_c^* , the consumption of Foreign buyers in DM meetings with type c sellers.

The changes in DM consumption quantities affect inflation rates, asset prices, and the nominal exchange rate as follows. First, from (15), the Home inflation rate μ falls. In equilibrium, the decrease in the nominal interest rate R_m leads to a decline in the real interest rate r_m . In other words, conventional easing has a liquidity effect on r_m . However, this liquidity effect does not fully offset the Fisher effect, resulting in a reduction in the inflation rate μ . In contrast, from (22), the Foreign inflation rate μ^* rises. This occurs because a decrease in R_m makes Foreign reserves more attractive relative to Foreign currency, given that the Foreign central bank keeps R_m^* at its target level. The reduction in the quantity of Foreign currency outstanding must be accompanied by an increase in μ^* in equilibrium.

From (50) and (51), the prices of Home bonds and interbank loans, z_b and z_e , both increase, while the price of Foreign bonds z_b^* remains unchanged. Also, from (43), the depreciation rate of the Home currency $\frac{\xi+1}{\xi}$ falls. These results are somewhat consistent with uncovered interest rate parity: a decrease in the Home bond yield relative to the Foreign bond yield partially leads to a current depreciation of the Home currency, followed by its future appreciation. These effects are summarized in Table 2.

6 Equilibrium with Active Interbank Market

In this section, we focus on cases where the nominal interest rate on Home bonds falls below the interest rate on Home reserves. This scenario occurs when the Home central bank substantially expands its balance sheet by exchanging reserves for Home bonds. Such an intervention increases the quantity of reserves—accessible only to retail banks—while reducing the quantity of Home bonds available to all financial intermediaries. As a result, the price of Home bonds rises relative to reserves, causing their nominal interest rate to fall below the interest rate on reserves.

When a nominal interest rate gap emerges between reserves and government bonds, banks create an additional type of asset through the interbank market: interbank loans. These loans effectively allow unconventional banks to hold reserves indirectly. Assuming that the conditions in Propositions 1-3 hold, two distinct scenarios arise in which Foreign banks borrow from unconventional banks in the interbank market.

<Home banks>		<Foreign banks>		<Unconventional banks>	
Asset	Liability	Asset	Liability	Asset	Liability
Home currency	Home currency	Foreign currency	Foreign currency	Foreign bonds	Foreign bonds
Home Reserves	Deposits	Reserves	Deposits	Home bonds	Home bonds
		Foreign bonds	Interbank borrowing	Interbank lending	Deposits

Figure 7: Bank asset portfolios in equilibrium (Case 2)

6.1 Case 2: Home bonds are used as collateral at the margin

If Home bonds are sufficiently scarce—but not extremely scarce—only unconventional banks hold Home bonds and use them, at the margin, to back their deposit liabilities. In this scenario, bank asset portfolios are illustrated in Figure 7. Home banks receive deposits and hold Home reserves and Home currency, while Foreign banks receive deposits and borrow in the interbank market to hold reserves, Foreign currency, and Foreign bonds. Unconventional banks receive deposits and acquire Home bonds, Foreign bonds, and interbank loans. Noting that the first-order conditions (24), (25), and (27) hold with equality, and that $\pi\gamma\hat{b}_u < b_u$ and $\pi(1-\gamma)\hat{b}_u^* = b_u^*$, we can derive the following four conditions same as those presented in Case 1.

$$(1-\delta)u'(x_d) + \delta = (1-\theta)u'(x_d^*) + \theta, \quad (45)$$

$$z_m = \frac{(1-\delta)u'(x_d) + \delta}{u'(x_c) - \delta u'(x_d) + \delta}, \quad (46)$$

$$z_m^* = \frac{u'(x_d^*)}{u'(x_c^*)}, \quad (47)$$

$$x_b^* = x_d^*. \quad (48)$$

The structure of asset prices can be expressed as:

$$\frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu} = z_m < z_e = \frac{\beta u'(x_d^*)}{\mu} = \frac{\beta u'(x_d^*)}{\mu} = \frac{\beta u'(x_b)}{\mu} = z_b, \quad (54)$$

$$\frac{\beta u'(x_d^*)}{\mu^*} = z_m^* = z_b^* = \frac{\beta u'(x_b^*)}{\mu^*}. \quad (51)$$

It shows that Home and Foreign banks strictly prefer Home reserves to Home bonds. It also suggests that retail banks may wish to engage in arbitrage by borrowing in the interbank market to increase their reserve holdings. However, Home banks do not do so because their balance sheets are tightly constrained by capital requirements, making arbitrage prohibitively costly. In contrast, Foreign banks actively borrow in the interbank market to earn arbitrage profits. Nevertheless, because they too face balance sheet constraints, the volume of arbitrage transactions is insufficient to eliminate the interest rate spread between Home reserves and interbank loans.

Similar to Case 1, all types of banks are subject to a consolidated balance sheet constraint due to the interconnectedness of asset markets across countries. From (1), (2), (5), (9), (13), (36)-(48), (51), and (54), noting that $b_h = b_f = b_h^* = e_h = 0$, the consolidated balance sheet constraint can be written as:

$$\begin{aligned}
& \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\
& + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\
& + (2-\alpha-\alpha^*)x_d^u \left[u'(x_d^u) + \frac{\theta}{1-\theta} \right] \\
& = (v+v^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] - \frac{\theta\omega}{(1-\theta)u'(x_d^*)}.
\end{aligned} \tag{55}$$

Equation (55) ensures that aggregate collateral demand equals its aggregate supply in equilibrium, with quantities adjusted to reflect the balance sheet costs faced by banks.²⁴

A necessary condition for the existence of this equilibrium is:

$$v - (2-\alpha-\alpha^*) \{ (1-\pi)x_d^u u'(x_d^u) + \pi\gamma x_b u'(x_b) \} < \omega \leq v - (2-\alpha-\alpha^*)\pi\gamma x_d^u u'(x_d^u), \tag{56}$$

Condition (56) indicates that the size of the Home central bank's balance sheet, ω , must be large enough for the limited supply of collateral in the financial system to induce unconventional banks to acquire interbank loans. At the same time, ω must be small enough to ensure a sufficiently large supply of Home bonds, allowing unconventional banks to use some bonds as collateral in equilibrium.

We can now analyze the effects of the Home central bank's monetary policy on the Home country and its transmission to the Foreign country.

6.1.1 Conventional easing: a fall in the nominal interest rate on reserves

Suppose the Home central bank lowers the nominal interest rate on reserves (i.e., raises z_m) without changing the size of its balance sheet ω . The qualitative effects of this intervention on DM

²⁴Similar to Case 1, we can solve for the consumption quantities in the DM meetings $(x_c, x_d, x_c^*, x_d^*, x_b, x_b^*, x_d^u)$, given monetary and fiscal policies $(z_m, z_m^*, \omega, \omega^*, v, v^*)$, using equations (45)-(48) and (54)-(55). Inflation rates are then determined by equations (15) and (22), asset prices by equations (51) and (54), and the nominal depreciation rate of the Home currency by equation (43).

x_c	x_d	x_c^*	x_d^*	x_b	x_b^*	x_d^u	μ	μ^*	$\frac{\xi+1}{\xi}$
↑	↓	↓	↓	↓	↓	↓	↓	↑	↓
z_m	z_e	z_b	$z_e - z_m$	$z_b - z_e$	z_b^*	$\mathcal{W}$	$\mathcal{W}^*$		
↑	↑	↑	↑	·	·	?	↓		

Table 3: Effects of an increase in z_m (Case 2)

consumption quantities and asset prices are the same as in Case 1. Specifically, from (45)-(48) and (54), the consumption quantities in all DM meetings supported by government bonds (x_b, x_b^*), collateral-backed deposits (x_d, x_d^*, x_d^u) and Foreign currency (x_c^*) decrease, whereas the consumption quantity in DM meetings involving Home currency (x_c) increases. From (15), (22), and (43)-(44), the Home inflation rate μ falls while the Foreign inflation rate μ^* rises, resulting in a decline in the depreciation rate of the Home currency $\frac{\xi+1}{\xi}$. Finally, from (51) and (54), the prices of Home bonds and interbank loans (z_b, z_e) both rise, whereas the price of Foreign bonds (z_b^*) remains unchanged.

What differs from the previous case is that, in this equilibrium, there is a price gap between Home reserves and interbank loans, and monetary policy can affect this gap. Using (45) and (54), the price difference between interbank loans and Home reserves, $z_e - z_m$, can be written as:

$$z_e - z_m = \frac{\beta\theta [u'(x_d^*) - 1]}{\mu}. \quad (57)$$

Thus, the price difference—or nominal interest rate gap—increases as z_m increases. These results are summarized in Table 3.

6.1.2 Unconventional easing: a balance sheet expansion

Suppose instead that the Home central bank increases the size of its balance sheet ω while keeping the nominal interest rate on reserves unchanged. Equation (55) shows that an increase in ω effectively decreases the supply of collateralizable assets. Then, from (45)-(48) and (54)-(55), all DM consumption quantities ($x_c, x_d, x_c^*, x_d^*, x_b, x_b^*, x_d^u$) decrease. The central bank expands its balance sheet by exchanging reserves for Home bonds. Since unconventional banks cannot hold reserves, this intervention tightens their collateral constraints, which in turn tightens the consolidated balance sheet constraint within the global financial system. Consequently, consumption declines in all DM meetings involving deposits backed by collateral (x_d, x_d^*, x_d^u). The tightening of balance sheet constraints also reduces the quantity of government bonds available as a means of payment, leading to lower consumption in those DM meetings (x_b, x_b^*).

A decrease in the supply of collateralizable assets raises their prices and lowers their real interest rates. However, the nominal interest rates on reserves are administratively set by central banks and are held fixed in this policy experiment. This implies that the decrease in the real interest rates on collateralizable assets must be accompanied by higher inflation rates— μ and μ^* —in equilibrium. Since the nominal interest rates on reserves and currencies remain constant, the fall in real interest rates only tightens the balance sheet constraints of retail banks, without affecting the relative prices

x_c	x_d	x_c^*	x_d^*	x_b	x_b^*	x_d^u	μ	μ^*	$\frac{\xi+1}{\xi}$
↓	↓	↓	↓	↓	↓	↓	↑	↑	↓
z_m	z_e	z_b	$z_e - z_m$	$z_b - z_e$	z_b^*	$\mathcal{W}$	$\mathcal{W}^*$		
·	↑	↑	↑	·	·	↓	↓		

Table 4: Effects of an increase in ω (Case 2)

of reserves to their respective currencies. As a result, the consumption of retail bank depositors in all DM meetings (x_c, x_d, x_c^*, x_d^*) decreases, as if experiencing a negative wealth effect.

Using (45) and (54), the price of interbank loans can also be expressed as:

$$z_e = \frac{z_m u'(x_d^*)}{(1 - \theta)u'(x_d^*) + \theta}, \quad (58)$$

which equals the price of Home bonds z_b . Since z_e rises as x_d^* decreases, an increase in ω raises the prices of both Home bonds and interbank loans in equilibrium. This occurs because of the decrease in the effective stock of collateral and the resulting tightening of balance sheet constraints. Thus, the price gap between these assets and reserves, $z_e - z_m$, also increases in equilibrium. Finally, from (44), (51), and (54), the nominal depreciation rate of the Home currency can be expressed as:

$$\frac{\xi+1}{\xi} = \frac{z_b^*}{z_b}. \quad (59)$$

Uncovered interest parity strictly holds in this case because the nominal depreciation rate depends only on the nominal interest rate differential between Home and Foreign bonds. Because z_b rises relative to z_b^* (equivalently, R_b falls relative to R_b^*), the depreciation rate falls, implying a current depreciation of the Home currency followed by an expected appreciation in the next period.

Interestingly, the welfare of Foreign bank depositors decreases because their consumption in DM meetings, x_c^* and x_d^* , both decrease. This decline occurs despite an increase in the nominal interest rate gap between interbank loans and Home reserves, which raises arbitrage profits for Foreign banks. Although Foreign banks use these higher profits to increase DM consumption for their depositors, that increase is outweighed by the reduction of consumption induced by the tightened collateral constraint. Therefore, x_c^* and x_d^* can decline even when Foreign banks earn greater arbitrage profits. These results are summarized in Table 4.

6.2 Case 3: Home bonds are used as a medium of exchange at the margin

In equilibrium, the price mechanism allocates each asset to the economic activity in which it is used most efficiently. When Home bonds become extremely scarce, they are allocated exclusively to certain DM meetings, where they serve as a medium of exchange. In this scenario, the composition of each bank's assets and liabilities remains the same as in the previous case. However, unlike before, unconventional banks no longer use Home bonds to back their deposit liabilities; instead, all Home bonds they hold are intended for withdrawal by depositors for use in DM transactions, as

<Home banks>		<Foreign banks>		<Unconventional banks>	
Asset	Liability	Asset	Liability	Asset	Liability
Home currency	Home currency	Foreign currency	Foreign currency	Foreign bonds	Foreign bonds
Home Reserves	Deposits	Reserves	Deposits	Home bonds	Home bonds
		Foreign bonds	Interbank borrowing	Interbank lending	Deposits

Figure 8: Bank asset portfolios in equilibrium (Case 3)

illustrated in Figure 8.

Noting that the first-order conditions (24), (25), and (27) hold with equality, and that $\pi\gamma\hat{b}_u = b_u$ and $\pi(1-\gamma)\hat{b}_u^* = b_u^*$, we can derive a set of equilibrium conditions identical to those in the previous two cases, such as equations (45)-(48) and (51). However, the price of Home bonds is higher than that of interbank loans:

$$\frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu} = z_m < z_e = \frac{\beta u'(x_d^u)}{\mu} = \frac{\beta u'(x_d^*)}{\mu} < \frac{\beta u'(x_b)}{\mu} = z_b. \quad (60)$$

As in Cases 1 and 2, the global financial system as a whole is subject to a consolidated balance sheet constraint. Using (1), (2), (5), (9), (13), (36)-(48), (51), and (60), and noting that $b_h = b_f = b_h^* = e_h = 0$, this consolidated balance sheet constraint can be expressed as:

$$\begin{aligned} & \alpha\rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\ & + \alpha^*\rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\ & + (2-\alpha-\alpha^*)(1-\pi\gamma)x_d^u \left[u'(x_d^u) + \frac{\theta}{1-\theta} \right] \\ & = \omega + v^* \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right]. \end{aligned} \quad (61)$$

In this equilibrium, an additional constraint is relevant for unconventional banks and DM meetings involving Home bonds:

$$(2-\alpha-\alpha^*)\pi\gamma x_b u'(x_b) = v - \omega. \quad (62)$$

x_c	x_d	x_c^*	x_d^*	x_b	x_b^*	x_d^u	μ	μ^*	$\frac{\xi+1}{\xi}$
↑	↓	↓	↓	·	↓	↓	↓	↑	↓
z_m	z_e	z_b	$z_e - z_m$	$z_b - z_e$	z_b^*	$\mathcal{W}$	$\mathcal{W}^*$		
↑	↑	↑	↑	?	·	?	↓		

Table 5: Effects of an increase in z_m (Case 3)

Equation (62) indicates that the effective quantity of Home bonds demanded as a medium of exchange equals the quantity supplied in the financial system. Since Home bonds are no longer part of the collateral, this condition uniquely determines the consumption quantity in DM meetings involving Home bonds x_b .²⁵

For this type of equilibrium to exist, the equilibrium solution must satisfy:

$$v - (2 - \alpha - \alpha^*)\pi\gamma x_d^u u'(x_d^u) < \omega \leq v, \quad (63)$$

Condition (63) indicates that the Home central bank's balance sheet ω must be sufficiently large so that Home bonds are extremely scarce and are used only as a medium of exchange in the financial system.

6.2.1 Conventional easing: a fall in the nominal interest rate on reserves

Suppose that the Home central bank decreases the nominal interest rate on reserves (or equivalently, increases z_m) while holding ω —the size of its balance sheet—constant. The effects of this intervention on DM consumption quantities are qualitatively similar to those in the previous case, except that, from (62), x_b is unaffected. Specifically, from (46), an increase in z_m leads to an increase in x_c and a decrease in x_d , reflecting a substitution of Home currency for Home reserves. Additionally, an increase in z_m drives up other asset prices and reduces the effective stock of collateral in equilibrium, as in the previous case. Due to the tightening of the consolidated balance sheet constraint, consumption quantities in DM meetings involving collateral-backed deposits (x_d, x_d^*, x_d^u) decrease.

Moreover, the increase in z_m relative to z_m^* raises the demand for Foreign reserves and bonds, inducing Foreign banks to substitute Foreign reserves for Foreign currency. As a result, the quantity of Foreign currency outstanding decreases, and the consumption quantity in Foreign currency-involved DM meetings x_c^* declines. Consumption in DM meetings involving Foreign bonds x_b^* also decreases, as the higher demand by Foreign banks reduces the amount of Foreign bonds available to unconventional banks.

As in the previous case, from (15) and (22), the Home inflation rate μ falls, while the Foreign inflation rate μ^* rises. Thus, from (43), the Home currency experiences current-period depreciation

²⁵The model can be solved as follows. First, equation (62) solves for x_b , and equations (45)-(48), (51), and (60)-(61) can be used to solve for all other DM consumption quantities $(x_c, x_d, x_c^*, x_d^*, x_b^*, x_d^u)$ given monetary and fiscal policies $(z_m, z_m^*, \omega, \omega^*, v, v^*)$. Then, equations (15) and (22) solve for inflation rates (μ, μ^*) , while equations (51) and (60) determine asset prices such as z_b, z_b^* , and z_e . Finally, equation (43) determines the nominal depreciation rate of the Home currency $\frac{\xi+1}{\xi}$.

x_c	x_d	x_c^*	x_d^*	x_b	x_b^*	x_d^u	μ	μ^*	$\frac{\xi+1}{\xi}$
$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$	$?$
z_m	z_e	z_b	$z_e - z_m$	$z_b - z_e$	z_b^*	$\mathcal{W}$	$\mathcal{W}^*$		
$\cdot$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\cdot$	$?$	$?$		

Table 6: Effects of an increase in ω (Case 3)

(due to a lower nominal interest rate on Home bonds relative to Foreign bonds) and is expected to appreciate in the next period. From (60), the prices of Home bonds and interbank loans, z_b and z_e , both rise, but the price difference between these two assets, $z_b - z_e$, is ambiguous. This is because a lower inflation rate tends to increase the price difference, while a higher price of interbank loans tends to decrease it. Finally, from (57), the price difference between interbank loans and reserves, $z_e - z_m$, increases. These results are summarized in Table 5.

6.2.2 Unconventional easing: a balance sheet expansion

Suppose instead that the Home central bank expands its balance sheet (i.e., increases ω) while keeping the nominal interest rate on reserves constant. In this equilibrium, an increase in ω entails a swap of collateralizable assets (Home reserves) for a means of payment in certain transactions (Home bonds). As a result, as shown in (61), this intervention increases the effective stock of collateral and relaxes the balance sheet constraint faced by banks.

This effect contrasts with the previous case, in which an increase in ω tightens the balance sheet constraint. In that earlier scenario, both Home bonds and reserves are used as collateral at the margin, and swapping reserves for bonds reduces the effective stock of collateral because reserves are a poor substitute for Home bonds as collateral. In the current setting, however, Home bonds are used only as a means of payment in certain DM transactions. Thus, the central bank's asset swap acts to increase the supply of collateralizable assets while reducing the supply of means of payment in those DM transactions. Specifically, from (62), consumption in DM meetings involving Home bonds x_b decreases, whereas from (45)-(48), (60), and (61), all other DM consumption quantities ($x_c, x_d, x_c^*, x_d^*, x_b^*, x_d^u$) increase.

An increase in the effective stock of collateralizable assets lowers their prices and relaxes the balance sheet constraint of Home banks. This works like a positive wealth effect, increasing their holdings of both Home currency and reserves. In equilibrium, the increase in the real quantity of currency holdings (and the associated increase in x_c) is accompanied by a decline in the Home inflation rate μ . The relaxed balance sheet constraint also allows Foreign banks to rebalance their portfolios by substituting Foreign currency for reserves in order to improve depositor welfare. For the real quantity of Foreign currency outstanding to increase in equilibrium, the Foreign inflation rate μ^* must fall.

From (58), the price of interbank loans z_e falls as x_d^* increases, implying that the price differential between interbank loans and Home reserves $z_e - z_m$ narrows. In contrast, from (60), the price of

Home bonds rises, causing the price differential between Home bonds and interbank loans $z_b - z_e$ to widen. The underlying reason is that, in this equilibrium, the market for Home bonds is effectively segmented from the market for reserves and other collateralizable assets. Consequently, an increase in ω —and the associated increase in the supply of collateral—tends to reduce the price of interbank loans z_e , while the reduced supply of Home bonds drives up their price z_b .

As inflation falls in both countries, the implied effect on the nominal depreciation rate of the Home currency, given by (43), is ambiguous. On the one hand, from (44), the nominal interest rate on Home bonds decreases (equivalently, z_b rises), which tends to depreciate the Home currency in the current period. On the other hand, the liquidity premium on Home bonds rises (equivalently, $u'(x_b)$ increases), which tends to appreciate the Home currency. As a result, the overall effect on the nominal depreciation rate is uncertain. This demonstrates that uncovered interest parity need not hold in this equilibrium: a decline in the nominal interest rate on Home bonds relative to Foreign bonds does not necessarily imply a nominal depreciation of the Home currency.

In contrast to the previous case—where the Home central bank’s balance sheet expansion reduces the welfare of Foreign bank depositors—their welfare increases in this equilibrium. The DM consumption quantities of Foreign bank depositors, x_c^* and x_d^* , both rise because Foreign banks benefit from the increase in the effective stock of collateral in the financial system. Although the nominal interest rate spread between interbank borrowing and reserves narrows, the relaxation of the consolidated balance sheet constraint allows Foreign banks to expand their holdings of currency and deposit liabilities. This, in turn, raises the corresponding DM consumption quantities and thereby improves the welfare of Foreign bank depositors. These results are summarized in Table 6.

7 Overnight Reverse Repurchase Agreement Facility

Due to limited arbitrage, a persistent spread emerged between the interest on reserve balances (IORB) rate and the federal funds (fed funds) rate. In response, the Federal Reserve introduced an overnight reverse repurchase agreement (ON-RRP) facility in September 2013 as a supplementary policy tool to improve control over the fed funds rate. This facility allows eligible counterparties to hold central bank liabilities overnight and earn interest—similar to reserve balances for depository institutions—while extending access to a broader range of entities, including banks, government-sponsored enterprises (GSEs), and money market mutual funds.

If the Federal Reserve supplies a sufficiently large quantity of ON-RRPs, the ON-RRP rate serves as a lower bound for the fed funds rate. If the fed funds rate were to fall below the ON-RRP rate, GSEs would prefer to invest in the ON-RRP facility rather than lend in the fed funds market. Thus, the introduction of the ON-RRP facility raises the fed funds rate and reduces arbitrage profits for foreign banks that rely on borrowing in the fed funds market. Consistent with this prediction, Figure 3 shows a downward shift in the relationship between the Federal Reserve’s balance sheet size and foreign banks’ share of total reserves following the introduction of the ON-RRP facility. In this section, we examine how the ON-RRP facility affects domestic and global financial markets.

We extend the model by allowing the Home central bank to issue interest-bearing ON-RRPs that can be held by unconventional banks. Like reserves and interbank loans, ON-RRPs can be used as collateral to back intermediaries' deposit liabilities, but they cannot serve as a medium of exchange for type b sellers. Each unit of ON-RRPs sells at price z_o in the CM and represents a claim to one unit of Home currency in the subsequent CM. Let ω denote the value of the Home central bank's liabilities (i.e., the size of its balance sheet), which now consist of currency, reserves, and ON-RRPs. Additionally, let $\bar{o}$ denote the real quantity of ON-RRPs held by the public, and define $\omega_o \equiv z_o \bar{o}$ as the value of ON-RRPs outstanding, so that $\omega \equiv \bar{c} + z_m \bar{m} + \omega_o$. We analyze the two cases considered in the previous section, in which Foreign banks actively borrow from unconventional banks in the interbank market.

7.1 Case 2: Home bonds are used as collateral at the margin

In the presence of an ON-RRP facility, the consolidated balance sheet constraint (55) can be rewritten as:

$$\begin{aligned}
& \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\
& + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\
& + (2-\alpha-\alpha^*)x_d^u \left[u'(x_d^u) + \frac{\theta}{1-\theta} \right] \\
& = (v+v^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] - \frac{\theta(\omega-\omega_o)}{(1-\theta)u'(x_d^*)}.
\end{aligned} \tag{64}$$

Equation (64) states that aggregate demand for collateral equals aggregate supply, with quantities adjusted to reflect the balance sheet costs faced by banks. Apart from (64), the equilibrium conditions are identical to those obtained in the absence of an ON-RRP facility, except that $z_o = z_e = z_b$ in equilibrium. The interest rates on ON-RRPs, interbank loans, and Home bonds are identical, rendering these assets perfect substitutes at the margin for unconventional banks.

Consider an experiment in which the Home central bank increases the size of the ON-RRP facility ω_o while holding constant both the size of its balance sheet ω and the nominal interest rate on reserves. Because ω is fixed, an increase in ω_o implies a decrease in the total value of currency and reserves outstanding, $\bar{c} + z_m \bar{m}$. This intervention relaxes the collateral constraints faced by unconventional banks and, in turn, relaxes the consolidated balance sheet constraint for the financial system as a whole. Consequently, from (45)-(48), (54), and (64), consumption quantities in all DM meetings increase.

The Home central bank's ON-RRP facility works in the opposite direction from a balance sheet expansion that involves exchanging reserves for Home bonds. As discussed in the previous section, such a balance sheet expansion can tighten the consolidated balance sheet constraint when limited access to reserves makes them less effective as collateral than the more widely available Home

bonds. By contrast, expanding the ON-RRP facility involves the central bank's swaps of ON-RRPs for reserves. Because unconventional banks can hold ON-RRPs as substitutes for interbank loans and Home bonds, the nominal interest rates on these assets rise until they align with the ON-RRP rate. This intervention therefore effectively relaxes the consolidated balance sheet constraint, enabling all types of banks to offer welfare-enhancing deposit contracts.

7.2 Case 3: Home bonds are used as a medium of exchange at the margin

In this case, the consolidated balance sheet constraint (61) can be rewritten as:

$$\begin{aligned}
& \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\
& + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\
& + (2-\alpha-\alpha^*)(1-\pi\gamma)x_d^u \left[u'(x_d^u) + \frac{\theta}{1-\theta} \right] \\
& = \omega + \frac{\theta\omega_o}{(1-\theta)u'(x_d^*)} + v^* \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right]. \tag{65}
\end{aligned}$$

Other equilibrium conditions are identical to those without an ON-RRP facility, except that $z_o = z_e$ in equilibrium. The ON-RRP rate equals the interest rate on interbank loans, making ON-RRPs perfect substitutes for interbank loans held by unconventional banks.

Suppose the Home central bank increases the size of the ON-RRP facility ω_o while holding constant both the size of its balance sheet and the nominal interest rate on reserves. From (45)-(48), (60), and (65), this intervention effectively increases the aggregate supply of collateral, leading to higher consumption in all DM meetings except those involving Home bonds. Because the market for Home bonds is effectively segmented from other markets, the central bank's exchange of ON-RRPs for reserves does not affect the real quantity of Home bonds outstanding, leaving consumption in DM meetings involving Home bonds x_b unchanged, as shown in (62).

In both cases considered above, replacing reserves with ON-RRPs increases welfare for all types of depositors—those at Home banks, Foreign banks, and unconventional banks. This result may appear counterintuitive, as retail banks hold fewer reserves, which might seem to limit their intermediation capacity. However, the reduction in reserve holdings is accompanied by a decline in Foreign banks' interbank borrowing and an increase in unconventional banks' ON-RRP holdings. These changes act to relax the collateral constraints faced by unconventional banks and lower the liquidity premium on all collateralizable assets, including reserves. Consequently, the real return on reserves rises, allowing both Home and Foreign banks to better serve their depositors despite holding fewer reserves.

7.3 Central bank balance sheet policies and retail banks' reserve holdings

As discussed in Section 2, the Federal Reserve's balance sheet expansion has been associated with an increase in the share of reserves held by foreign banks. This pattern persists even after the

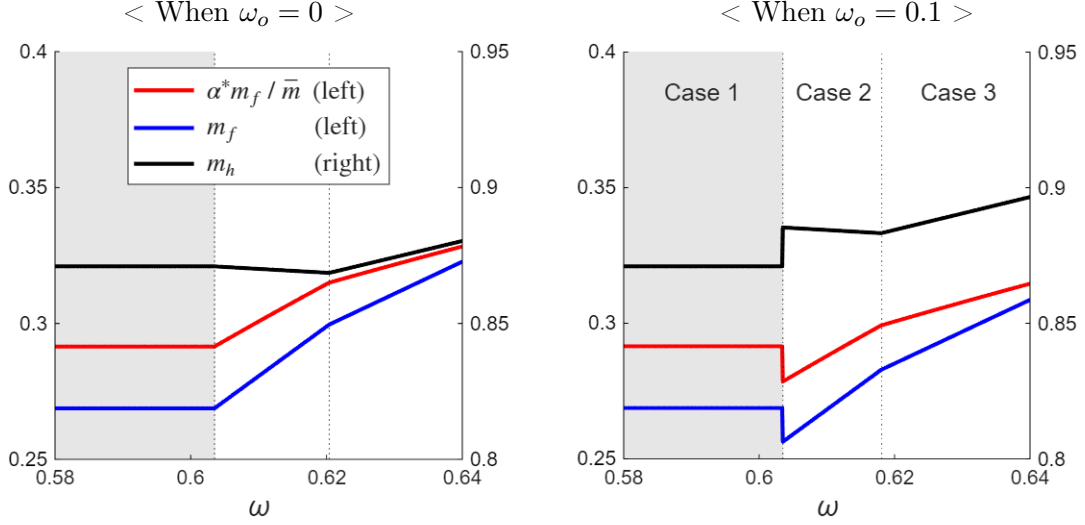


Figure 9: Reserve holdings of each bank type and Foreign banks' reserve share

introduction of the ON-RRP facility, although the facility shifts down the level of foreign banks' reserve share (see Figure 3).

Our model delivers predictions that are consistent with these empirical observations. Figure 9 illustrates how retail banks' reserve holdings respond to changes in the size of the central bank's balance sheet before and after the introduction of the ON-RRP facility.²⁶ In both scenarios, when the size of the Home central bank's balance sheet ω becomes sufficiently large (Case 2), a further increase in ω raises Foreign banks' share of reserves, $\alpha^*m_f/\bar{m}$, by increasing their reserve holdings, m_f , while reducing those of Home banks, m_h . Even when ω becomes very large (Case 3), an increase in ω continues to raise Foreign banks' share of reserves, as m_f increases by more than m_h .

The introduction of the ON-RRP facility lowers the share of reserves held by Foreign banks at all levels of ω . This result mirrors the empirical pattern in Figure 3, where the ON-RRP facility shifts down the relationship between the size of the Federal Reserve's balance sheet and foreign banks' share of reserves. In the model, this occurs because ON-RRPs reduce arbitrage activity by Foreign banks, leading to a decline in their demand for reserves. From the perspective of unconventional banks, ON-RRPs are close substitutes for interbank loans and therefore relax their collateral constraints. The resulting decline in asset prices—or, equivalently, increase in real interest rates—allows Home banks, which do not engage in arbitrage, to increase their reserve holdings.

8 Conclusion

Considering how monetary policy works in the United States since late 2008, it has become essential to understand the implications of foreign banks' behavior in interbank markets for U.S. monetary

²⁶In this numerical example, we assume that $u(x) = \frac{x^{1-\sigma}}{1-\sigma}$, $\sigma = 0.45$, $\beta = 0.96$, $\rho = 0.2$, $\pi = \gamma = 0.5$, $\alpha = 0.45$, $\alpha^* = 0.6$, $\delta = 0.3$, $\theta = 0.25$, $v = 0.8$, $v^* = 0.85$, $z_m = 0.98$, $z_m^* = 0.97$, and $\omega^* = 0.2$.

policy. To explore this, we have constructed a two-country model of retail and unconventional banks that explicitly incorporates institutional differences across financial intermediaries. The model generates active interbank markets—where foreign retail banks borrow from unconventional banks—and an interest rate spread between reserves and interbank loans, consistent with the data.

When foreign banks actively engage in arbitrage in interbank markets, the effects of the Federal Reserve’s balance sheet expansion depend critically on the stock of government bonds held by the public. In particular, such an expansion can reduce global welfare. This adverse effect arises when the Federal Reserve injects relatively poorer collateral (reserves accessible only to retail banks) while taking out relatively superior collateral (government bonds that are widely accessible) from the financial system. As a result, balance sheet constraints tighten for all financial intermediaries, leading to contractions in economic activity and welfare losses for both domestic and foreign consumers.

We have also analyzed the global implications of introducing an overnight reverse repurchase agreement (ON-RRP) facility. By allowing unconventional banks to earn interest on the Federal Reserve’s overnight liabilities, the ON-RRP facility has particularly strong effects on foreign banks, given their reliance on interbank borrowing. Although the facility reduces foreign banks’ arbitrage profits, an expansion of the ON-RRP facility—accompanied by a contraction in reserves—relaxes balance sheet constraints throughout the global financial system. Consequently, both domestic and foreign consumers experience welfare gains.²⁷

²⁷Another experiment we consider in the model is a central bank’s foreign exchange (FX) intervention. In particular, an FX intervention by the *foreign* central bank—implemented through swaps of (its domestic) *foreign* reserves for (its foreign) *home* government bonds—has effects that are qualitatively similar to those of the *home* central bank’s balance sheet expansion. In both cases, the intervention increases the effective stock of collateral in the global financial system while reducing the supply of *home* government bonds available for certain transactions. Additional details are provided in Appendix A.2.

References

- Afonso, Gara, Roc Armenter, and Benjamin R. Lester. 2019. “A Model of the Federal Funds Market: Yesterday, Today, and Tomorrow.” *Review of Economic Dynamics* 33:177–204.
- Afonso, Gara, Gonzalo Cisternas, Brian Gowen, Jason Miu, and Joshua Younger. 2023a. “Who’s Borrowing and Lending in the Fed Funds Market Today?” *Federal Reserve Bank of New York Liberty Street Economics* 20231010. URL <https://libertystreeteconomics.newyorkfed.org/2023/10/whos-borrowing-and-lending-in-the-fed-funds-market-today/>.
- Afonso, Gara, Alex Entz, and Eric LeSueur. 2013. “Who’s Borrowing in the Fed Funds Market?” *Federal Reserve Bank of New York Liberty Street Economics* 20131209. URL <https://libertystreeteconomics.newyorkfed.org/2013/12/whos-borrowing-in-the-fed-funds-market/>.
- Afonso, Gara, Domenico Giannone, Gabriela La Spada, and John C. Williams. 2023b. “Scarce, Abundant, or Ample? A Time-Varying Model of the Reserve Demand Curve.” *Federal Reserve Bank of New York Staff Reports* 1019:1–64.
- Afonso, Gara, Gabriela La Spada, Thomas M. Mertens, and John C. Williams. 2023c. “The Optimal Supply of Central Bank Reserves under Uncertainty.” *Federal Reserve Bank of New York Staff Reports* 1077:1–35.
- Afonso, Gara and Ricardo Lagos. 2015. “Trade Dynamics in the Market for Federal Funds.” *Econometrica* 83 (1):263–313.
- Anderson, Alyssa, Wenxin Du, and Bernd Schlusche. 2025. “Arbitrage Capital of Global Banks.” *Journal of Finance* 80 (5):2591–2638.
- Anderson, Alyssa and Dave Na. 2024. “The Recent Evolution of the Federal Funds Market and its Dynamics during Reductions of the Federal Reserve’s Balance Sheet.” *FEDS Notes* URL <https://doi.org/10.17016/2380-7172.3548>.
- Armenter, Roc and Benjamin R. Lester. 2017. “Excess Reserves and Monetary Policy Implementation.” *Review of Economic Dynamics* 23:212–235.
- Arrata, William, Benoît Nguyen, Imène Rahmouni-Rousseau, and Miklos Vari. 2020. “The Scarcity Effect of QE on Repo Rates: Evidence from the Euro Area.” *Journal of Financial Economics* 137 (3):837–856.
- Backus, David K. and Patrick J. Kehoe. 1989. “On the Denomination of Government Debt. A Critique of the Portfolio Balance Approach.” *Journal of Monetary Economics* 23 (3):359–376.
- Banegas, Ayelen and Manjola Tase. 2022. “Reserve Balances, the Federal Funds Market and Arbitrage in the New Regulatory Framework.” *Journal of Banking and Finance* 118:1–14.
- Bech, Morten L. and Elizabeth Klee. 2011. “The Mechanics of a Graceful Exit: Interest on Reserves and Segmentation in the Federal Funds Market.” *Journal of Monetary Economics* 58 (5):415–431.
- Bianchi, Javier and Saki Bigio. 2022. “Banks, Liquidity Management and Monetary Policy.” *Econometrica* 90 (1):391–454.

- Cavalcanti, Ricardo de O. and Neil Wallace. 1999. “Inside and Outside Money as Alternative Media of Exchange.” *Journal of Money, Credit and Banking* :443–457.
- Diamond, Douglas W. 1984. “Financial Intermediation and Delegated Monitoring.” *The Review of Economic Studies* 51 (3):393–414.
- Diamond, Douglas W. and Philip H. Dybvig. 1983. “Bank Runs, Deposit Insurance, and Liquidity.” *Journal of Political Economy* 91 (3):401–419. URL <http://www.journals.uchicago.edu/doi/10.1086/261155>.
- DiSalvo, James. 2019. “How Foreign Banks Changed After Dodd-Frank.” *Federal Reserve Bank of Philadelphia Economic Insights* 3:1–6.
- Du, Wenxin. 2023. “Foreign Banking Organizations in the United States and the Price of Dollar Liquidity.” *Federal Reserve Bank of New York Liberty Street Economics* January 11. URL <https://libertystreeteconomics.newyorkfed.org/2023/01/foreign-banking-organizations-in-the-united-states-and-the-price-of-dollar-liquidity/>.
- Ennis, Huberto M. 2018. “A Simple General Equilibrium Model of Large Excess Reserves.” *Journal of Monetary Economics* 98:50–65.
- Ennis, Huberto M. and Todd Keister. 2008. “Understanding Monetary Policy Implementation.” *FRB Richmond Economic Quarterly* 94 (3):235–263.
- Ennis, Huberto M. and Alexander L. Wolman. 2015. “Large Excess Reserves in the United States: A View from the Cross-Section of Banks.” *International Journal of Central Banking* 11 (1):251–289.
- Geromichalos, Athanasios and Kuk Mo Jung. 2018. “An Over-the-Counter Approach to the FOREX Market.” *International Economic Review* 59 (2):859–905.
- Geromichalos, Athanasios and Ina Simonovska. 2014. “Asset Liquidity and International Portfolio Choice.” *Journal of Economic Theory* 151:342–380.
- Gissler, Stefan and Borghan Narajabad. 2017. “The Increased Role of the Federal HomeLoan Bank System in Funding Markets, Part 1: Background.” *FEDS Notes* URL <https://doi.org/10.17016/2380-7172.2070>.
- Gomis-Porqueras, Pedro, Timothy Kam, and Junsang Lee. 2013. “Money, Capital, and Exchange Rate Fluctuations.” *International Economic Review* 54 (1):329–353.
- Gomis-Porqueras, Pedro, Timothy Kam, and Christopher Waller. 2017. “Nominal Exchange Rate Determinacy under the Threat of Currency Counterfeiting.” *American Economic Journal: Macroeconomics* 9 (2):256–273.
- Gu, Chao, Fabrizio Mattesini, Cyril Monnet, and Randall Wright. 2013. “Banking: A New Monetarist Approach.” *Review of Economic Studies* 80 (2):636–662.
- Head, Allen and Shouyong Shi. 2003. “A Fundamental Theory of Exchange Rates and Direct Currency Trades.” *Journal of Monetary Economics* 50 (7):1555–1591.
- Herrenbrueck, Lucas. 2017. “An Open-Economy Model with Money, Endogenous Search, and Heterogeneous Firms.” *Economic Inquiry* 55 (4):1648–1670.

- Kim, Duhyeong. 2023. “International Effects of Quantitative Easing and Foreign Exchange Intervention.” *Journal of International Economics* 145:1–35.
- Kreicher, Lawrence L., Robert N. McCauley, and Patrick McGuire. 2014. “The 2011 FDIC Assessment on Banks’ Managed Liabilities: Interest Rate and Balance-Sheet Responses.” In *Taxation and Regulation of the Financial Sector*.
- Lagos, Ricardo and Randall Wright. 2005. “A Unified Framework for Monetary Theory and Policy Analysis.” *Journal of Political Economy* 113 (3):463–484.
- Lee, Seungduck and Kuk Mo Jung. 2020. “A Liquidity-Based Resolution of the Uncovered Interest Parity Puzzle.” *Journal of Money, Credit and Banking* 52:1397–1433.
- Matsuyama, Kiminori, Nobuhiro Kiyotaki, and Akihiko Matsui. 1993. “Toward a Theory of International Currency.” *The Review of Economic Studies* 60 (2):283–307.
- Oguri, Junko and Cristoforo Pizzimenti. 2025. “Ample Reserves for Whom? The Role of Foreign Banks in U.S. Monetary Policy Implementation.” :1–103URL <http://dx.doi.org/10.2139/ssrn.5656410>.
- Poole, William. 1968. “Commercial Bank Reserve Management in a Stochastic Model: Implications for Monetary Policy.” *Journal of Finance* 23 (5):769–791.
- Rocheteau, Guillaume and Randall Wright. 2005. “Money in Search Equilibrium, in Competitive Equilibrium, and in Competitive Search Equilibrium.” *Econometrica* 73 (1):175–202.
- Stern, Elisheva R. 2022. “Bank Portfolio Choice and Monetary Policy Transmission in the Face of a New Federal Funds Market.”
- Williamson, Stephen D. 1986. “Costly Monitoring, Financial Intermediation, and Equilibrium Credit Rationing.” *Journal of Monetary Economics* 18 (2):159–79.
- . 2019. “Interest on Reserves, Interbank Lending, and Monetary Policy.” *Journal of Monetary Economics* 101:14–30.
- . 2023. “Interest Rate Control and Interbank Markets.” *Working Paper* :1–33.
- Wright, Randall and Alberto Trejos. 2001. “International Currency.” *Topics in Macroeconomics* 1 (1):153460051013.
- Zhang, Cathy. 2014. “An Information-Based Theory of International Currency.” *Journal of International Economics* 93 (2):286–301.

A Appendix

A.1 Borrowing and Lending in the Federal Funds Market

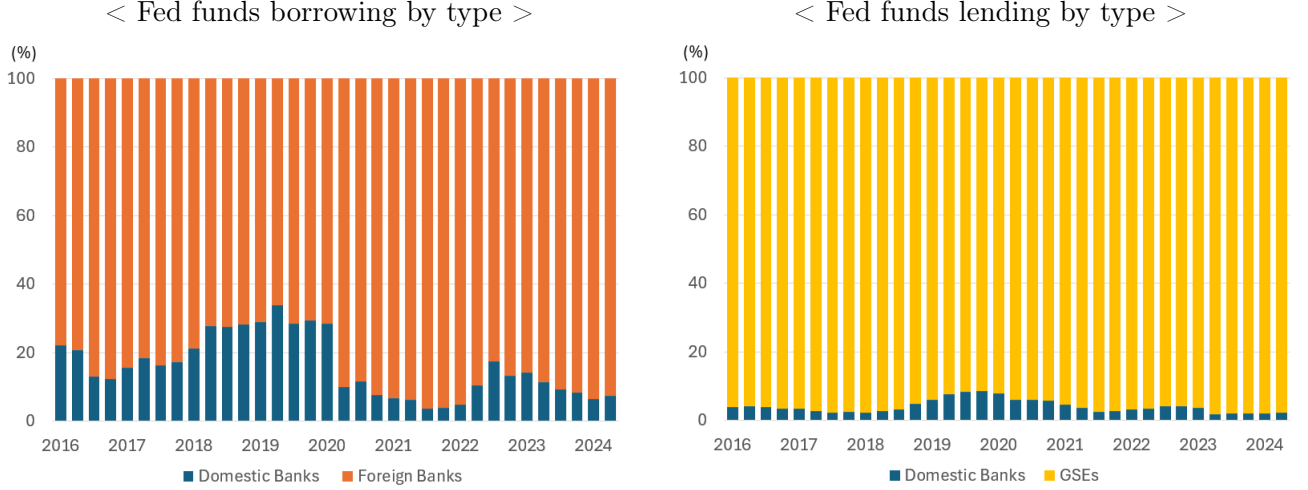


Figure 10: Fed funds market share by type

Source: [Anderson and Na \(2024\)](#)

A.2 Foreign Central Bank's Foreign Exchange Intervention

When the Federal Reserve began purchasing large quantities of U.S. Treasury securities in late 2008, many other central banks intervened in foreign exchange (FX) markets by purchasing foreign currency-denominated assets, primarily U.S. Treasury securities. Between 2009 and 2014, while the Federal Reserve acquired approximately \$2.0 trillion worth of U.S. Treasury securities, other central banks increased their holdings of U.S. dollar-denominated FX reserves by about \$1.7 trillion.²⁸ What are the effects of the central banks' FX interventions on the U.S. economy and the global financial system? To address this issue, we extend the model developed in Sections 3–6.

In Section 3, we assumed that the quantity of Home bonds held by the Foreign central bank—denoted by b_{FC} —was zero. Here, we relax this assumption and allow the Foreign central bank to hold Home bonds to influence the exchange rate between Foreign and Home currencies. Specifically, suppose the Foreign central bank determines the value of its holdings of Home bonds, denoted by $\chi \equiv z_b b_{FC}$, in addition to its other policy instruments. The Foreign fiscal policy rule can then be rewritten as:

$$v^* = \bar{c}^* + z_m^* \bar{m}^* + z_b^* \bar{b}^* - \chi.$$

The Foreign central bank's demand for Home bonds must also be incorporated into the market-

²⁸Data on U.S. dollar-denominated FX reserves are from the IMF's Currency Composition of Official Foreign Exchange Reserves (<https://data.imf.org/?sk=e6a5f467-c14b-4aa8-9f6d-5a09ec4e62a4>). Between 2009 and 2014, total FX reserves in the world increased from \$7.4 trillion to \$11.6 trillion, while officially reported U.S. dollar-denominated reserves rose from \$2.7 trillion to \$4.4 trillion. During this period, however, the currency composition of approximately 42–43% of total FX reserves was unreported. This implies that the actual increase in U.S. dollar-denominated FX reserves may be larger than the reported amount.

clearing condition, so we replace equation (37) with:

$$\alpha b_h + \alpha^* b_f + (2 - \alpha - \alpha^*) b_u + b_{FC} = \bar{b},$$

To capture the behavior of foreign banks in the U.S. interbank market, we focus on the two cases analyzed in Section 6, in which Foreign banks actively borrow from unconventional banks in the interbank market.

A.2.1 Case 2: Home bonds are used as collateral at the margin

In this case, somewhat surprisingly, the equilibrium conditions with $b_{FC} > 0$ are identical to those with $b_{FC} = 0$. That is, the equilibrium is characterized by (45)-(48) and (54)-(55), and Foreign central bank's FX interventions have no effect on equilibrium prices or consumption quantities. The intuition behind this result is as follows. When the Foreign central bank purchases Home bonds, it swaps Foreign reserves for Home bonds. This supply-side shift initially raises the price of Home bonds relative to Foreign reserves. In response, unconventional banks substitute interbank loans for Home bonds, which puts downward pressure on the price of Home bonds and upward pressure on the price of interbank loans. Foreign banks then increase interbank borrowing to acquire additional Foreign reserves, which puts upward pressure on the price of Foreign reserves and downward pressure on the price of interbank loans. As a result, despite changes in the composition of assets in the financial system, all asset prices remain unchanged because the increased activity in the interbank market fully offsets the initial supply shift.

An alternative interpretation relies on the fact that both Home bonds and Foreign reserves are used as collateral at the margin. Although these assets are held by different types of banks, the collateral value (or pledgeability) of Home bonds for unconventional banks is equivalent to that of Foreign reserves for Foreign banks. Consequently, the Foreign central bank's swap of Foreign reserves for Home bonds does not alter the effective aggregate stock of collateral in the global financial system, as interbank market activities fully absorb the change in the composition of collateral.

The Foreign central bank's purchases of Home bonds therefore merely increase lending and borrowing in the interbank market, without affecting equilibrium prices or consumption quantities. This result is reminiscent of Ricardian equivalence theorem, which states that changes in the timing of taxation only alter private saving and public borrowing without affecting equilibrium prices or consumption quantities. [Backus and Kehoe \(1989\)](#) similarly demonstrate the irrelevance of sterilized FX interventions for equilibrium prices and quantities in an open-economy cash-in-advance model. Although our framework generates home bias in asset portfolios—unlike [Backus and Kehoe \(1989\)](#)—FX interventions remain irrelevant in the presence of an interbank market.²⁹

²⁹FX interventions may matter when Foreign reserves are imperfect substitutes for Home bonds. In particular, if the pledgeability of Home bonds as collateral for unconventional banks differs from that of Foreign reserves for Foreign banks—for example, due to differences in sovereign default risk—FX interventions can affect equilibrium outcomes. [Kim \(2023\)](#) examines the implications of such differential asset pledgeability for the effects of FX interventions.

A.2.2 Case 3: Home bonds are used as a medium of exchange at the margin

Given the Foreign central bank's holdings of Home bonds $b_{FC} > 0$, the consolidated balance sheet constraint (61) can be rewritten as:

$$\begin{aligned}
& \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\
& + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\
& + (2-\alpha-\alpha^*)(1-\pi\gamma)x_d^u \left[u'(x_d^u) + \frac{\theta}{1-\theta} \right] \\
& = \omega + (v^* + \chi) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right].
\end{aligned} \tag{66}$$

In addition, the condition determining consumption in DM meetings involving Home bonds, 62, becomes:

$$(2-\alpha-\alpha^*)\pi\gamma x_b u'(x_b) = v - \omega - \chi. \tag{67}$$

All other equilibrium conditions—(15), (22), (43), (45)-(48), and (60)—remain unchanged.

What happens if the Foreign central bank increases χ (the value of its Home bond holdings) while keeping the size of its balance sheet unchanged? From (66)-(67) and other equilibrium conditions, this FX intervention has effects that are qualitatively identical to those of an increase in ω (the Home central bank's balance sheet expansion). In both cases, the intervention increases the effective stock of collateral available in the global financial system while reducing the quantity of Home bonds available for certain DM transactions. Specifically, (67) implies that consumption in DM meetings requiring Home bonds x_b decreases, whereas (45)-(48), (60), and (66) imply that all other DM consumption quantities ($x_c, x_d, x_c^*, x_d^*, x_b^*, x_d^u$) increase. All other effects are consistent with those reported in Table 6.

The qualitative equivalence between the Foreign central bank's purchases of Home bonds and the Home central bank's balance sheet expansion—which also involves Home bond purchases—stems from the substitutability between Home and Foreign reserves. Because Foreign banks hold both types of reserves as collateral at the margin, the two interventions affect their behavior in the same way. Quantitatively, however, the effects differ. An increase in χ has a larger impact on the aggregate stock of collateral than an increase in ω . This difference reflects variation in pledgeability between Home and Foreign reserves: although both serve as collateral for Foreign banks, local currency-denominated Foreign reserves are more pledgeable than foreign currency-denominated Home reserves. As a result, the Foreign central bank's swap of Foreign reserves for Home bonds increases the effective stock of collateral by more than the Home central bank's swap of Home reserves for Home bonds.

A.3 Omitted Proofs

Proof of Proposition 1: First, we prove that $b_h^* = m_h^* = 0$ and $b_u^* = \pi(1-\gamma)\hat{b}_u^*$ when constraints (5), (9), and (13) bind in equilibrium. Suppose these constraints bind, but $b_h^* > 0$. Then, first-order

conditions (17), (19), (24), (26), and (32) imply that

$$z_b^* = \frac{\beta\{(1-\delta)[(1-\theta)u'(x_d) + \theta] + \delta\}}{\mu^*} = \frac{\beta u'(x_b^*)}{\mu^*} = \frac{\beta u'(x_d^*)}{\mu^*} = z_m^*.$$

It follows that, using (21) and (27),

$$\begin{aligned} z_e &\leq \frac{\beta u'(x_d^*)}{\mu} \\ &= \frac{\beta\{(1-\delta)[(1-\theta)u'(x_d) + \theta] + \delta\}}{\mu} \\ &< \frac{\beta[(1-\delta)u'(x_d) + \delta]}{\mu} \\ &\leq z_e, \end{aligned}$$

which is a contradiction. Therefore, it must be the case that $b_h^* = 0$ when capital and collateral constraints bind in equilibrium. We omit the proof of $m_h^* = 0$ since it is straightforward.

Suppose instead that constraints (5), (9), and (13) bind, but $b_u^* > \pi(1-\gamma)\hat{b}_u^*$. From (24), (26), (32), and (34), we obtain the following equilibrium condition:

$$z_b^* = \frac{\beta u'(x_b^*)}{\mu^*} = \frac{\beta[(1-\theta)u'(x_d^u) + \theta]}{\mu^*} \geq \frac{\beta u'(x_d^*)}{\mu^*} = z_m^*.$$

Then, it must be the case that, using (27) and (35),

$$\begin{aligned} z_e &= \frac{\beta u'(x_d^u)}{\mu} \\ &> \frac{\beta[(1-\theta)u'(x_d^u) + \theta]}{\mu} \\ &\geq \frac{\beta u'(x_d^*)}{\mu} \\ &\geq z_e, \end{aligned}$$

which is a contradiction. Thus, in equilibrium, $b_u^* = \pi(1-\gamma)\hat{b}_u^*$ given binding capital and collateral constraints.

Given that binding constraints lead to $b_h^* = m_h^* = 0$ and $b_u^* = \hat{b}_u^*$, we can show a sufficient condition under which constraints (5), (9), and (13) bind in equilibrium. Consider cases where the quantity of collateralizable assets is sufficiently large to support an equilibrium with $b_h \geq 0$, $b_f \geq 0$, $b_f^* \geq 0$, $m_h \geq 0$, $m_f \geq 0$, $m_f^* \geq 0$, and $b_u \geq \pi\gamma\hat{b}_u$. Using (1), (2), and (15)-(42), we can rewrite the balance sheet constraints (5), (9), and (13) as the following single inequality:

$$\begin{aligned}
& v + \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)}\right] v^* \\
& \geq \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \alpha(1-\rho)x_d \left[u'(x_d) + \frac{\delta}{1-\delta} \right] \\
& \quad + \alpha^* \rho x_c^* u'(x_c^*) \left[1 + \frac{\theta}{(1-\theta)u'(x_d^*)} \right] + \alpha^*(1-\rho)x_d^* \left[u'(x_d^*) + \frac{\theta}{1-\theta} \right] \\
& \quad + (2-\alpha-\alpha^*) \left\{ (1-\pi)x_d^u u'(x_d^u) + \pi\gamma x_b u'(x_b) + \pi(1-\gamma)x_b^* \left[u'(x_b^*) + \frac{\theta}{1-\theta} \right] \right\}.
\end{aligned}$$

The above equation indicates that the aggregate demand for collateral to support consumption in DM meetings (the right-hand side) cannot be greater than its aggregate supply (the left-hand side) in equilibrium. Suppose that banks' balance sheet constraints do not bind in equilibrium. Then, banks will be able to issue a sufficient quantity of deposit liabilities to support an efficient level of consumption in DM meetings with type d sellers. That is, $x_d = x_d^* = x_d^u = x^e$ where $u'(x^e) = 1$. This implies that from (16), (24), (29) and (32), it must be the case that $x_b = x_b^* = x^e$. Then, the above inequality can be rewritten as:

$$\begin{aligned}
& v + \frac{v^*}{1-\theta} \\
& \geq \alpha \rho x_c \left[u'(x_c) + \frac{\delta}{1-\delta} \right] + \frac{\alpha^* \rho x_c^* u'(x_c^*)}{1-\theta} \\
& \quad + x^e \left[\frac{\alpha(1-\rho)}{1-\delta} + \frac{\alpha^*(1-\rho)}{1-\theta} + \frac{(2-\alpha-\alpha^*)[1-\theta(1-\pi-\pi\gamma)]}{1-\theta} \right] \\
& > x^e \left[\frac{\alpha(1-\rho)}{1-\delta} + \frac{\alpha^*(1-\rho)}{1-\theta} + \frac{(2-\alpha-\alpha^*)[1-\theta(1-\pi-\pi\gamma)]}{1-\theta} \right]
\end{aligned}$$

Therefore, if v and v^* are sufficiently small so that

$$(1-\theta)v + v^* < x^e \left\{ (1-\rho) \left[\frac{\alpha(1-\theta)}{1-\delta} + \alpha^* \right] + (2-\alpha-\alpha^*)[1-\theta(1-\pi+\pi\gamma)] \right\},$$

banks' balance sheet constraints—(5), (9), and (13)—bind in equilibrium.

Proof of Proposition 2: First, note that from (35), unconventional banks are indifferent between borrowing and lending in the interbank market. From the first-order conditions for Home and Foreign banks ((20), (21), (27), and (28)), the equilibrium price of interbank loans must satisfy the following:

$$\begin{aligned}
(1-\delta)u'(x_d) + \delta & \leq \frac{\mu z_e}{\beta} \leq u'(x_d), \\
(1-\theta)u'(x_d^*) + \theta & \leq \frac{\mu z_e}{\beta} \leq u'(x_d^*).
\end{aligned}$$

A necessary condition for Foreign banks to borrow in the interbank market while Home banks remain inactive is given by:

$$u'(x_d^*) < u'(x_d).$$

Foreign banks have an incentive to borrow in the interbank market only if they hold Home reserves. Given that both Home and Foreign banks hold Home reserves, i.e., $m_h > 0$ and $m_f > 0$, the following condition must hold in equilibrium (18 and 25):

$$(1 - \delta)u'(x_d) + \delta = (1 - \theta)u'(x_d^*) + \theta.$$

Therefore, the above necessary condition can be rewritten as:

$$\delta > \theta.$$

Proof of Proposition 3: From (2), (32), (33), and (40), the quantity of Foreign bonds held by Foreign banks can be expressed as:

$$b_f^* = \frac{\mu^*[v^* - \omega^* - (2 - \alpha - \alpha^*)(1 - \gamma)\pi x_b^* u'(u_b^*)]}{\alpha^* \beta u'(x_b^*)}.$$

For b_f^* to be positive, it must be the case that

$$v^* > \omega^* + (2 - \alpha - \alpha^*)(1 - \gamma)\pi x_b^* u'(u_b^*).$$

In this case, the prices of Home bonds, Foreign bonds, and interbank loans can be expressed, respectively, as:

$$\begin{aligned} z_m = z_b &= \frac{\beta[(1 - \delta)u'(x_d) + \delta]}{\mu} = \frac{\beta[(1 - \theta)u'(x_d^*) + \theta]}{\mu}, \\ z_m^* = z_b^* &= \frac{\beta u'(x_b^*)}{\mu^*} = \frac{\beta u'(x_d^*)}{\mu^*}, \\ z_e &= \begin{cases} \frac{\beta[(1 - \theta)u'(x_d^*) + \theta]}{\mu} & \text{if } b_h > 0 \\ \frac{\beta u'(x_d^*)}{\mu} & \text{if } b_h = 0 \end{cases} \end{aligned}$$

Therefore, in equilibrium, all the asset prices are positively correlated.